

Semantic Audit: An Elementary Comparison for Four-Arc Regions in a Planar Strip Density

Independent semantic review completed and accepted by
openai-codex/gpt-6-astra

Verdict and scope

Verdict: equivalent. The author handoff and assembled mathematical bodies faithfully explain the four frozen roots. The primary conclusion is the corrected explicit-family comparison: for every real $\lambda > 1$, the constant γ_λ is positive, and for every $0 < h \leq 1$ there is a real r satisfying

$$0 < r < h, \quad A_\lambda(E_r) = A_\lambda(C_h), \quad P_\lambda(C_h) - P_\lambda(E_r) > \frac{\gamma_\lambda}{h} > 0.$$

The separately named scalar comparison, the complete elementary route, and unconditional geometric realization are also present with their frozen hypotheses and quantifier order. Neither equal area nor an area/perimeter bridge is assumed by the primary theorem. [[@lean:CMVElementary.geometric_comparison](#)] [[@lean:CMVElementary.scalar_comparison](#)] [[@lean:CMVElementary.elementary_route](#)] [[@lean:CMVElementary.geometric_realization](#)]

The review covers every one of the 506 inventoried non-generated declarations, both author prose fields, all declaration statements and explanations, and both assembled Markdown manuscripts. The supplied original note is a source to be corrected, not the literal theorem being certified. The unrestricted classification corollary remains explicitly informal and dependent on cited external results.

Findings

- No critical mismatch was found in the frozen root statements, their informal translations, or the mathematical proof route.
- No critical main-proof or supplement error was found. The renderer-owned title, credit, repository locator, section scaffolding and HTML Lean-reference blocks are not author-body defects. In particular, the empty generated-family section is renderer scaffolding, not evidence of a missing family.
- No external citation issue was found: both citations identify supplied fixed documents and specific mathematical locators.
- The review approves the corrected explicit-family theorem, not either erroneous area-minus-perimeter display in the original PDF, a general bounded-variation formalization, or a kernel proof of the external classification.

Evidence and elementary proof route

The checked statements preserve the strict density hypothesis, the closed right endpoint of the four-arc curvature interval, and the strictly smaller positive competitor parameter. The margin is the specified density-dependent constant divided by curvature; no positive lower bound uniform as $\lambda \downarrow 1$ is asserted. The scalar functions remain total real definitions, while geometric and differential claims use their stated domains. The symbol j denotes the explicitly defined continuous extension, not an endpoint differentiation operation. [[@lean:CMVElementary.ScalarStatement](#)] [[@lean:CMVElementary.GeometricStatement](#)] [[@lean:CMVElementary.jprime](#)]

The radical normalization is proved for positive density. Interior differentiation gives

$$S'(z) = -2D(z), \quad J'(u) = j(u), \quad H(u) = J(u) - uj(u).$$

Evenness and strict increase of D determine the signs of j . Equal endpoint values give the lower bound for J . On the closed lower third, the actual proof of the lower bound for H uses the decreasing-derivative tangent lemma with independently proved premises. It does not project the desired bound from the record it is constructing. On the remaining interval it uses $H \geq J$. The manuscript correctly distinguishes that dependency from the separately established derivative and limit identities for H . `[@lean:CMVElementary.DerivativePilot.sqrt_normalization]` `[@lean:CMVElementary.hasDerivAt_S]` `[@lean:CMVElementary.hasDerivAt_J]` `[@lean:CMVElementary.MonotoneTangent.h_lower_third_of_monotone_tangent]` `[@lean:CMVElementary.H_lower]` `[@lean:CMVElementary.auxiliary_claims]`

The algebraic same-curvature identities yield

$$\begin{aligned} A_4(h) - A_3(h) &\geq \gamma_\lambda/h^2 > 0, \\ [P_4(h) - hA_4(h)] - [P_3(h) - hA_3(h)] &\geq \gamma_\lambda/h > 0. \end{aligned}$$

The variational identity is transported by neighborhood equality with the normalized scalar library, rather than by equality at a single point. Divergence of the three-arc area chooses a positive left endpoint. The continuous equal-area level set in a compact interval has a greatest element. Its maximality and the intermediate value theorem give the strict area barrier to its right. No root uniqueness or global area monotonicity is used. `[@lean:CMVElementary.same_curvature_claims]` `[@lean:CMVElementary.three_arc_eventuallyEq]` `[@lean:CMVElementary.three_arc_variational_hasDerivAt]` `[@lean:CMVElementary.A3_tendsto_atRight_zero]` `[@lean:CMVElementary.exists_greatest_level_root]`

For the same selected root and fixed target area, the support expression satisfies

$$Q(t) = P_3(t) + t[A_4(h) - A_3(t)], \quad Q'(t) = A_4(h) - A_3(t) > 0 \quad (r < t < h).$$

Continuity on the closed comparison interval gives $Q(r) < Q(h)$. Combining this strict improvement with the non-strict same-curvature lower bound gives the strictly greater perimeter margin in the scalar root. Differentiation remains inside the interval even when $h = 1$. `[@lean:CMVElementary.greatest_root_claims]` `[@lean:CMVElementary.strict_support_comparison]` `[@lean:CMVElementary.scalar_comparison]`

Model and geometry fidelity

The area is the Lebesgue integral of the piecewise constant density. The complete-frontier perimeter is determined by Euclidean one-dimensional Hausdorff measure. The coordinate product type is transported to its Euclidean realization before that Hausdorff measure is formed; the default maximum product metric is not used for arc length. The interfaces $y = \pm 1$ have density one. The generic perimeter functional accepts a measure, but the final geometric roots instantiate it with the canonical measure of the specified carrier. `[@lean:StripDensity]` `[@lean:WeightedArea]` `[@lean:EuclideanPlane]` `[@lean:FrontierMeasure]` `[@lean:WeightedPerimeter]`

The four-arc and three-arc sets are defined by literal disk, half-plane and horizontal-section inequalities. Exact carrier equalities connect these definitions to the integral infrastructure. The four-arc lower cap is the horizontal-axis reflection of the upper cap and lies in $y \leq -1$; its non-endpoint exterior arc lies strictly below that interface. Strict attaching-chord points are interior to the assembled regions, whereas the entire three-arc bottom segment remains exposed frontier. `[@lean:CMVElementary.GeometryBridge.C_assembly]` `[@lean:CMVElementary.GeometryBridge.E_assembly]` `[@lean:CMVElementary.GeometryBridge.reflected_cap_below]` `[@lean:OneSidedCircularCap.arcPoint_in_open_side]` `[@lean:CMVElementary.GeometryBridge.four_upper_chord]` `[@lean:CMVElementary.GeometryBridge.four_lower_chord]` `[@lean:CMVElementary.GeometryBridge.three_strict_chord]` `[@lean:CMVElementary.GeometryBridge.E_bottom_frontier]`

The full cap-angle range retains major exterior caps for $r < 1/2$. At $r = 1/2$, the carrier is exactly the radius-two disk centered at $(0, 1)$, and the bottom segment is a singleton of zero length, not a deleted or pinched branch. The four-arc endpoint $h = 1$ is included; its selected competitor still satisfies $r < 1$. Closedness, boundedness, measurability, density integrability and finite frontier mass are conclusions of the realization proofs.

[@lean:CMVElementary.ThreeArcRealization] [@lean:CMVElementary.GeometryBridge.E_half_disk]
 [@lean:TypeThreeAssembly.sideHalfWidth_eq_zero_iff]
 [@lean:TypeThreeAssembly.exact_euclidean_bottomSegmentCarrier_hausdorffMeasure]
 [@lean:CMVElementary.GeometryBridge.three_regularity] [@lean:CMVElementary.GeometryBridge.four_regularity]

The geometric area formulas are actual carrier integral identities. The canonical perimeter formulas use exact frontier decompositions, actual Euclidean arc and segment lengths, and null pairwise junction overlaps. Auxiliary modeled or parametrized perimeter sums are explicitly bridged to those integrals before use. The bottom interface contributes its full density-one segment length. The realization records are constructed from the density and curvature bounds and are not supplied as hypotheses to evade those bridges. [@lean:FourArcAssembly.weightedArea_formula]
 [@lean:fourArc_frontier_weightedPerimeter_eq] [@lean:TypeThreeAssembly.weightedArea_formula]
 [@lean:TypeThreeAssembly.weightedPerimeter_eq_modeledWeightedPerimeter]
 [@lean:CMVElementary.geometric_realization]

Inventory, manuscript and certificate coverage

The declaration inventory contains 75 entries from `CMVElementary.lean`, 52 from `CMVElementaryGeometry.lean`, 140 from `CMVGeometry.lean`, 20 from `FourArcCandidate.lean`, 80 from `FrontierPerimeter.lean`, two from `GreatestLevelRoot.lean`, 14 from `LeanSuffixAnalytic.lean`, three from `MorgansArcFunction.lean`, and 120 from `TypeThreeAssembly.lean`. Every entry has exactly one review below. Each review copies the author's statement verbatim and uses the inventory's declaration-code digest, not the whole-file digest. All nine inventoried source-file digests and all 506 declaration-code digests were checked against the retained Lean files.

The configured generated-glob list is empty, the author supplies no generated families, and no inventory declaration is marked generated. The generated-family review list is therefore correctly empty. There are no unexamined indexed representatives hidden behind a family summary.

A direct manuscript consistency check matched all 506 supplement statements, explanations, source locations, declaration hashes and Lean code blocks to the handoff and frozen source slices. After removing only renderer-owned reference blocks and handoff reference tokens, the assembled main body matches the handoff. The supplement introduction also matches. The mathematical-body check found no unmatched inline or display delimiters, unbalanced braces, mismatched environments, dollar delimiters, unresolved Lean tokens or exposed-notation candidates. It checked 925 inline expressions, 605 displays and 81 Lean-reference tokens. Manual reading additionally checked the theorem-proof order, consequential displays, notation, endpoint explanations, closing scope and bibliography. This is a review of the supplied Markdown content, not a claim to have visually certified newly generated PDF pagination.

The retained Lean gate reports successful build and exact inhabited-root checks, with only `propext`, `Classical.choice` and `Quot.sound` in the four root axiom reports. The retained scalar and geometric audit commands report successful exits and inspect required proof dependencies and forbidden older comparison routes. Their serialized standard output is flagged as truncated; this review does not claim to have read a complete transitive dependency dump. The audit source, recorded outcomes and the actual scalar and geometric proof terms support the stated route. No new Lean compilation was performed in this semantic pass. Compilation evidence was used only for the formal statements and dependencies, not as a substitute for the statement-by-statement semantic comparisons.

External-source checks

Cañete, Miranda Jr. and Vittone, *Some isoperimetric problems in planes with density*, Journal of Geometric Analysis **20** (2010), 243–290, is identified by the supplied `Canete2010.pdf` and the article's arXiv identifier. The inspected copy is the preprint `arXiv:0906.1256v1`; its SHA-256 is

0ee9b4787aa3c09f9a4ed05fb66563a820cd13596af401b76e7900e597711230. The cited locators are substantive and exact: Section 2 for reduced-boundary perimeter; Lemma 3.8 and its proof, Step 2, for the candidate families and horizontal symmetry of the four-arc case; Proposition 3.9 for vertical reflection symmetry of arbitrary minimizers; Conjecture 3.12 for the proposed classification; and Theorem 3.16 for the remaining area ranges and transition coexistence. The latter leaves the four-arc alternative, which the corrected explicit-family theorem excludes after the cited identification.

The supplied note by Skilyn Leon, Rosa Pavlak, Evelyn Pulla and Xi Sisi Shen is identified by `CMV_Strip_Density_Problem.pdf`, its two Theorem 1 statements, Sections 3–6 and Remarks 1–3. Its checked SHA-256 is `d04a2e8e2ad45fe505345037c046ceaeb2a8d63b9e2b1c1eeb020b5251c576ff`, exactly the frozen original. The original Theorem 1 on page 2 and equation (3.2) on page 3 print the erroneous area-minus-perimeter expression. The handoff explicitly corrects both to perimeter minus perimeter and leaves the source PDF unchanged. Page 3 was also rendered and inspected to resolve the text-extraction warning; it explicitly states the agreement of complete and reduced boundaries up to a length-null set for its piecewise smooth Jordan regions. The new manuscript correctly acknowledges that this boundary statement was already in the draft.

The informal corollary does not conflate complete-frontier perimeter with the perimeter of arbitrary area-equivalent representatives. For these explicit piecewise-circular regions, agreement with reduced-boundary perimeter is a conventional geometric transfer; for arbitrary minimizers, the cited classification and regular-representative results must first be invoked. This transfer and the external classification are not extra Lean hypotheses or axioms. No uniqueness of minimizers is inferred, and transition coexistence is retained.

Disposition

Accept the author handoff and the assembled mathematical bodies as equivalent to the frozen package claim. No mathematical, citation or author-body format revision is required by this pass. Acceptance is limited to the corrected explicit-family theorem and its three named supporting roots; the cited classification corollary remains informal, and the unchanged source note’s two false printed displays remain explicitly uncertified.

Package-level findings

Relation: equivalent

No package-level semantic errors were reported.

Declaration-by-declaration audit

AdmissibleCompetitor

Source hash: 52c3c4597213ca5cc7d4e12b21ad852ff148bcb8179b3f52a9c03b7d3665311b

Relation: equivalent

Source `CMVGeometry.lean`: 3952–3960: All four datatype alternatives are listed, and none is interpreted as a minimization assertion.

Findings: none.

AdmissibleCompetitor.WeightedArea

Source hash: d58b51a28690771d1b404e9048905bb6eae43f08f6798b27166b1234f3251f5e

Relation: equivalent

Source `CMVGeometry.lean`: 3968-3975: The area accessor uses zero in the empty branch and the corresponding actual integral accessor in each other constructor branch.

Findings: none.

AdmissibleCompetitor.WeightedPerimeter

Source hash: 17b5c5914d38bc0479d502c48cde9715c303aa7070585ade87ba56d307321437

Relation: equivalent

Source `CMVGeometry.lean`: 3976-3980: Every constructor's perimeter is computed from its carrier's canonical Euclidean frontier; no branch can select an independent boundary measure.

Findings: none.

AdmissibleCompetitor.carrier

Source hash: b57d4c4c33776a05ffbbbc22f1b363a7f356aa7cd089d7c2b3335ead1bb65644

Relation: equivalent

Source `CMVGeometry.lean`: 3961-3967: The carrier accessor follows each constructor exactly, returning the empty set only in the empty branch.

Findings: none.

CMVElementary.A3

Source hash: cb2c3a767813dd6743a002f60c9affd33c258de379854d6c91d5deadc49363b5

Relation: equivalent

Source `CMVElementary.lean`: 41-44: The affine shape substitution and numerator coefficient agree, with the squared parameter denominator unchanged.

Findings: none.

CMVElementary.A3_continuousAt

Source hash: 2aa80bdfce7c015108ca71987762c2d6eec97c1a5537e77e8a4add60919387ba

Relation: equivalent

Source `CMVElementary.lean`: 636-639: Only a nonzero parameter is assumed, and the assertion is continuity at that point for any real density.

Findings: none.

CMVElementary.A3_tendsto_atRight_zero

Source hash: bbfeb34d411d80b4748ecb5317b26152556ee012a4c270e7d6f639c2f722c14a

Relation: equivalent

Source `CMVElementary.lean`: 667-678: The proof transports the right-hand divergence by eventual equality; the author states the same one-sided infinite limit.

Findings: none.

CMVElementary.A4

Source hash: 4a8eaff3edf0f9c1b5bdc4eebb83b5e656403bb3492c00db3ddccd5bd349b4df

Relation: equivalent

Source CMVElementary.lean:49-50: The factor two and the curvature-weighted radical term reproduce the formal scalar area exactly.

Findings: none.

CMVElementary.AuxiliaryClaims

Source hash: d28b69e2724973e27afe7c069b33cc5b996e8abf00ea05832fa32bd7ac02f146

Relation: equivalent

Source CMVElementary.lean:74-102: Every record field is translated, including closed-interval continuity, open-interval derivatives, the lower-third sign claims and the right-hand limit.

Findings: none.

CMVElementary.B

Source hash: 5258c909f795e5bbd4f5ef3c79d0f8a12d32481ff10a7612a85a63a34f2572c3

Relation: equivalent

Source CMVElementary.lean:30-32: The definition retains both inverse-trigonometric terms and their density scaling; total real arguments are explicit.

Findings: none.

CMVElementary.C

Source hash: dac24abe77e8fbb5c839aa4c94d56e4b1ad024651bcedbea970677ac66fb909e

Relation: equivalent

Source CMVElementaryGeometry.lean:39-41: The formal union includes the literal core, upper cap and its reflected image; no measure-based surrogate replaces the carrier.

Findings: none.

CMVElementary.D

Source hash: 03723a810a1a63abdd42a17f0c33c9ae90ceffff8b93ca60d7527ed4f150d055

Relation: equivalent

Source CMVElementary.lean:33-35: The two radicands and their subtraction order coincide; nonnegativity is not made part of the definition.

Findings: none.

CMElementary.D_abs

Source hash: 5db30368ee603248cbd1e4396a9f85e0f017b44b1517e201076738ac74691b68

Relation: equivalent

Source CMElementary.lean: 474-478: The absolute-value identity is an unconditional consequence of evenness, with all real arguments retained.

Findings: none.

CMElementary.D_even

Source hash: 7b4f3c46b671c85fa9365ac233355bdd565a25b839e1a28c1a0f55338b48441a

Relation: equivalent

Source CMElementary.lean: 391-393: Evenness is genuinely unconditional for all real parameters because only squared arguments occur.

Findings: none.

CMElementary.D_nonneg

Source hash: 8ceb0e3455b90fa64499eee81dc63f1d0e17f21da82cc4b48255006f6cae2e2a

Relation: equivalent

Source CMElementary.lean: 394-403: The author preserves the stronger all-real argument domain of the nonnegativity theorem and the strict density hypothesis.

Findings: none.

CMElementary.D_strictMono

Source hash: 6591c7dbdbd88b546f03ea069e9476a8557083684fc0bb6944048ffcd4f3f44c

Relation: equivalent

Source CMElementary.lean: 404-409: Strict increase on the closed interval is obtained from positive interior derivatives and continuity; the endpoint is not dropped.

Findings: none.

CMElementary.DerivativePilot.D_continuous

Source hash: c1e2565224862417f88cc26d21ac89a1eb14395b552065dfde479c911555d2da

Relation: equivalent

Source CMElementary.lean: 283-286: Global continuity for every real density matches the total square-root definition, not merely the geometric domain.

Findings: none.

CMElementary.DerivativePilot.D_differentiableOn

Source hash: ea05176def925de59cfb082583ec3ac8045ebc1e63e39e642102b8aba520a3fd

Relation: equivalent

Source CMElementary.lean: 234-239: The formal differentiability domain is the open radical interval, exactly as translated.

Findings: none.

CMElementary.DerivativePilot.D_normalization

Source hash: 042c0b88a8b4aad471bbb7cd59d649270eaf82feb3afdd1eee7df1acb25b044d

Relation: equivalent

Source CMElementary.lean:171-176: The statement is equality of whole real functions, obtained by pointwise square-root normalization, not merely equality on the geometric interval.

Findings: none.

CMElementary.DerivativePilot.H_continuous

Source hash: 1e9b7d4f89a28deb3482239f3babe2259c9d36603e39a28f27e59937b9c67228

Relation: equivalent

Source CMElementary.lean:291-294: The assertion is global continuity of the total expression; it does not assert endpoint differentiability.

Findings: none.

CMElementary.DerivativePilot.H_limit_zero

Source hash: 3fa895803ef1aea4d3d273718ce23287763c81a52827fb5d35f0d8e9245453cf

Relation: equivalent

Source CMElementary.lean:300-312: The right-hand limit is unconditional and follows from continuity and the zero-endpoint value, not from endpoint differentiation.

Findings: none.

CMElementary.DerivativePilot.J_zero

Source hash: 89ec878363f839a318ed701d106d998cde8ae845aaa0b897db75b6d68f34219d

Relation: equivalent

Source CMElementary.lean:295-299: The endpoint evaluation is unconditional in the real density parameter, matching the reflection simplification in the proof.

Findings: none.

CMElementary.DerivativePilot.deriv_D

Source hash: 30a180dad9754ac8b27ff3cc498a89ca981d4205c49ae9ee5cd3a6be7fbe5a18

Relation: equivalent

Source CMElementary.lean:204-209: This is the derivative-value consequence of the preceding derivative theorem, with identical hypotheses and no endpoint extension.

Findings: none.

CMElementary.DerivativePilot.deriv_D_neg

Source hash: fd9ec3f8436fd5a4cbfa39412991f7fc77c1e1d9d9d6557a16b9878e3343ac1c

Relation: equivalent

Source CMElementary.lean: 222-227: The negative argument interval is preserved, and multiplication by the positive coefficient yields the claimed negative derivative.

Findings: none.

CMElementary.DerivativePilot.deriv_D_pos

Source hash: 2ecd442f21def60181fa9ad69af687a0bd5040c9880fdb5da7b9948bb3ba3df7

Relation: equivalent

Source CMElementary.lean: 228-233: The positive argument interval is preserved; the later monotonicity statement is not confused with this local derivative assertion.

Findings: none.

CMElementary.DerivativePilot.deriv_H_pos_of_J

Source hash: 303bde784351003abd78c85956bcd7a5a548511d46fc0a1e12378795be4dcb69

Relation: equivalent

Source CMElementary.lean: 276-282: The conditional derivative premise is retained and the strict positivity follows from the preceding negative derivative combination.

Findings: none.

CMElementary.DerivativePilot.derivative_coefficient_pos

Source hash: 3b1dd20767a51d60e6a3184ceb9440736e6ef81eb3d05b79b085bff9fa38ccf9

Relation: equivalent

Source CMElementary.lean: 210-221: Positive reciprocal comparison is correctly stated even at the zero argument; no unnecessary nonzero-argument hypothesis is added.

Findings: none.

CMElementary.DerivativePilot.hasDerivAt_D

Source hash: 25ed9e8fe57ff5b6872ee31ca5e64c1daca4f7b0b2c2e6d3d202561278ec1daa

Relation: equivalent

Source CMElementary.lean: 187-203: The exact derivative coefficient and its density factor agree, with differentiability asserted only at interior arguments.

Findings: none.

CMElementary.DerivativePilot.hasDerivAt_H_of_J

Source hash: 7aab8b2b2fb04f8525ea011add719fb8e7540c817e8f26fb8a7543159fe02613

Relation: equivalent

Source CMElementary.lean: 266-275: The derivative premise for the auxiliary function is explicitly retained; the product-rule cancellation does not silently discharge it.

Findings: none.

CMElementary.DerivativePilot.hasDerivAt_jprime

Source hash: 94ad44e520ab302a768e15871418b7bd98703e11200ad83d0d006b7c265ad1c9

Relation: equivalent

Source CMElementary.lean:253-265: The chain-rule factors agree, and the author retains the full open unit interval rather than restricting to the lower third.

Findings: none.

CMElementary.DerivativePilot.jprime_continuous

Source hash: 793ddb37fb86b542cee82ad8c3620a2966360336486c29f8d5e41082b4022fff

Relation: equivalent

Source CMElementary.lean:287-290: The affine composition preserves global continuity for every real density, as the author states.

Findings: none.

CMElementary.DerivativePilot.jprime_deriv_neg

Source hash: 58ff7311ad822a152063bce0332ab5d88a9004d4d249e43b945f5276cfd42ce7

Relation: equivalent

Source CMElementary.lean:244-252: The two derivative arguments have opposite signs, and the coefficients give the strict inequality in the stated direction.

Findings: none.

CMElementary.DerivativePilot.lower_third_signed_arguments

Source hash: 25037fe9d96454915fdfae797c1768e12148be94cb81c1f09cd91ca928805616

Relation: equivalent

Source CMElementary.lean:240-243: Both strict interval memberships follow from the same lower-third hypothesis and are stated without an irrelevant density assumption.

Findings: none.

CMElementary.DerivativePilot.radical_denominators

Source hash: 11268e29fea0051e68c749ca17d7b30526b0cc0f472498c62a0360246108b38e

Relation: equivalent

Source CMElementary.lean:177-186: All four strict positivity conclusions and the open argument interval are retained; these justify the derivative denominators.

Findings: none.

CMVElementary.DerivativePilot.sqrt_normalization

Source hash: 4eb056d0891d93b9a10938ae816157b2fdb95e7a6c30ea9942c36548da144840

Relation: equivalent

Source CMVElementary.lean:165-170: The formal hypothesis is positive density, not density exceeding one; the author preserves validity for every real argument.

Findings: none.

CMVElementary.E

Source hash: f1d34fbfd5ea8d77eb2e384dd1e6f5674c85b88684e3f0510bf703608445bde2

Relation: equivalent

Source CMVElementaryGeometry.lean:54-57: The region is the union of the actual core and cap; the bottom segment is a later frontier conclusion, not extra input data.

Findings: none.

CMVElementary.ElementaryRoute

Source hash: 4b48fa17e2a280d7cb3e3056c212a5e018edffc49f5b8e8a39a8fc2035f440f6

Relation: equivalent

Source CMVElementary.lean:155-164: The conjunction retains all four ingredient records and the universally quantified existence of greatest-root witnesses.

Findings: none.

CMVElementary.FourArcRealization

Source hash: b4e3701b509f6d802066e8cb6919211e45a8c1e2c32ad1ab317a1b5e812217d1

Relation: equivalent

Source CMVElementaryGeometry.lean:80-96: The existential candidate bridge, physical incidence, all regularity and integral fields, both strict chord interiors and reflected-cap location are preserved.

Findings: none.

CMVElementary.GeometricRealizationStatement

Source hash: bf9d5e6f83903980043feb1686756fd6224652d95175a5fa7761171f2e85ae57

Relation: equivalent

Source CMVElementaryGeometry.lean:117-127: The universal realization contract includes the four-arc endpoint but excludes it for the three-arc family, exactly as stated.

Findings: none.

CMVElementary.GeometricStatement

Source hash: 1f7b139bbebd695d7241107d88e880fde926dba2dd2619d65eb42a12b0005fa8

Relation: equivalent

Source CMVElementaryGeometry.lean:58-69: The quantified proposition uses canonical area and complete-frontier perimeter with the corrected perimeter-minus-perimeter conclusion.

Findings: none.

CMVElementary.GeometryBridge.C_area

Source hash: 917e6af80fc905a99360aafda85c3da61226b43930c862e06a404c527ac58964

Relation: equivalent

Source CMVElementaryGeometry.lean: 719-737: The actual carrier integral is evaluated via the candidate area theorem with a proved physical incidence relation.

Findings: none.

CMVElementary.GeometryBridge.C_assembly

Source hash: 36111eea65048699c328fdfad74c7928894d6d24d25d7d1aeac3df275ad3b185

Relation: equivalent

Source CMVElementaryGeometry.lean: 639-643: The three component set equalities identify the full four-arc coordinate carrier with the assembly carrier.

Findings: none.

CMVElementary.GeometryBridge.C_perimeter

Source hash: 8c97261adbc31a2bd30879f7c2a5ab32572dd45c6907559d577bd3fdcf08d3b2

Relation: equivalent

Source CMVElementaryGeometry.lean: 738-747: The candidate perimeter already uses the canonical frontier; its angle formula simplifies to the stated scalar perimeter.

Findings: none.

CMVElementary.GeometryBridge.E_area

Source hash: 26911f7fec1c4fdf0821ee4fddb03a1b3f0fb250f807d52386163b0c1a116545

Relation: equivalent

Source CMVElementaryGeometry.lean: 345-357: The actual carrier equality and weighted-area theorem are rewritten into the article's scalar formula with the same parameter.

Findings: none.

CMVElementary.GeometryBridge.E_assembly

Source hash: c1f0d7af54ce95a3387b069cc3722450d390ba0144860c4a288a74d575a7dd2c

Relation: equivalent

Source CMVElementaryGeometry.lean: 224-228: The core and cap equalities give equality of the entire carriers, which is the prerequisite for transporting the integrals.

Findings: none.

CMElementary.GeometryBridge.E_bottom_frontier

Source hash: ed0fb53bddfd325b31955a995791b4c29602ba5684e3bd105c2d1aa514d54853

Relation: equivalent

Source CMElementaryGeometry.lean:288-304: Closedness gives the frontier inclusion and the assembly bottom-segment theorem gives the reverse inclusion, including zero width.

Findings: none.

CMElementary.GeometryBridge.E_bottom_section

Source hash: cb328b960a0430a84de9eaba267689fedbc4342a75a8851f06d00f004a71af25

Relation: equivalent

Source CMElementaryGeometry.lean:262-287: The section identity is unconditional in both real parameters; the bottom radicand vanishes even under total division conventions.

Findings: none.

CMElementary.GeometryBridge.E_half_disk

Source hash: 33bc8668cf575aec788fdfae9213ec87b05ab1165d2368372477c264572d4074

Relation: equivalent

Source CMElementaryGeometry.lean:229-248: The disk equality is proved for every real density, not only the physical density range; both pieces have the same center at this parameter.

Findings: none.

CMElementary.GeometryBridge.E_perimeter

Source hash: a49183f07902e3ede19ad8a842f68990ab55b78592388d3d979e3541ccf7970b

Relation: equivalent

Source CMElementaryGeometry.lean:358-370: The formal identity concerns the canonical complete frontier; the angle and sine rewrites preserve the bottom-segment contribution.

Findings: none.

CMElementary.GeometryBridge.K3_core

Source hash: 2864cc2e686087b540a455347a5834437db804c051a3d2801e42dcb5bd3e614c

Relation: equivalent

Source CMElementaryGeometry.lean:176-212: The proof gives literal equality of coordinate sets using the nonnegative strip radicand and the section decomposition, not merely equal area.

Findings: none.

CMElementary.GeometryBridge.K_core

Source hash: a3ce44dbe7cbd338db626634d2aa7957319833c03c7387021e4daeb815f196b9

Relation: equivalent

Source CMElementaryGeometry.lean:583-597: Substitution in the general core-width theorem produces the literal radical offset without adding incidence or area assumptions.

Findings: none.

CMElementary.GeometryBridge.abs_le_sqrt_iff

Source hash: 2c06f5120bf21413fd7a76c0fdf930961207a98f7083bc089c888386262367b5

Relation: equivalent

Source CMElementaryGeometry.lean:140-150: The nonnegative-radical hypothesis is retained in both directions of the squaring equivalence.

Findings: none.

CMElementary.GeometryBridge.cap3_outer

Source hash: 2140a5194c99cf5b4856225e98d69256f9c79b99ca0647ae5ad4c1adf63bf5da

Relation: equivalent

Source CMElementaryGeometry.lean:213-223: Expansion of the cap center and radius yields exact set equality on the whole assembly domain, including major caps.

Findings: none.

CMElementary.GeometryBridge.cap4_lower

Source hash: 72a1ca1b6ff442b2e011b6c32059f321f81f287d8c74261f2c7a82c1ad2b72a7

Relation: equivalent

Source CMElementaryGeometry.lean:611-638: Both directions of the reflected-image equality are proved; the lower cap is not confused with a region merely below the upper interface.

Findings: none.

CMElementary.GeometryBridge.cap4_upper

Source hash: f441242c0d9f6318efed4cd2ba6e4ad6a81fcdc8e3d7771eacd9f798d79f5bc2

Relation: equivalent

Source CMElementaryGeometry.lean:598-610: The common-radius theorem identifies the cap disk exactly, with matching upper half-plane and center.

Findings: none.

CMElementary.GeometryBridge.cap_bounded

Source hash: 5f789f5ecfe21a6d9d280eafb5b4f8cb77643861d589df8496f30b5949a88aea

Relation: equivalent

Source CMElementaryGeometry.lean:407-409: The cap carrier is a subset of its bounded coordinate disk, with the constructor supplying admissibility.

Findings: none.

CMElementary.GeometryBridge.core_bounded

Source hash: f823fe31b5fbe1402665006d839f3f12aa85b1c8db11bbfdf640c49c7cbfe890

Relation: equivalent

Source CMElementaryGeometry.lean:644-660: The proof bounds the rectangle and both disk-contained side caps, giving boundedness of their union.

Findings: none.

CMElementary.GeometryBridge.core_width

Source hash: af8d9f8e6e0bf124954fc510fa0ca2b2c3e4d56c02f7c034bd8fff1b46ccb71f

Relation: equivalent

Source CMElementaryGeometry.lean:539-582: The proof identifies the actual core carrier with its width sections using the chord geometry; the author preserves the absolute height bound.

Findings: none.

CMElementary.GeometryBridge.disk_bounded

Source hash: e6037bd412d0410bbc35975c588a201fc2bbeccbaafb12028d4c97e43ab9925

Relation: equivalent

Source CMElementaryGeometry.lean:390-406: The boundedness statement permits any real radius symbol because its square determines the disk; the author does not add positivity.

Findings: none.

CMElementary.GeometryBridge.fourCandidate

Source hash: 9592404e2a8e9db2de9f06d6cd4a116fe2bca9a8fee984888822274385f8c fb9

Relation: equivalent

Source CMElementaryGeometry.lean:371-379: The constructed candidate has exactly the prescribed curvature and inverse-cosine angle, and its constructor bounds follow from the stated hypotheses.

Findings: none.

CMElementary.GeometryBridge.fourCandidate_snell

Source hash: d7b2da8df52c4eb799f1fd88604ac20c38b7b681ea3f936138fce2ab8263e2ba

Relation: equivalent

Source CMElementaryGeometry.lean:380-389: Both density jump and incidence are established for the constructed candidate by the valid cosine-of-inverse-cosine identity.

Findings: none.

CMElementary.GeometryBridge.four_euclidean_frontier_finite

Source hash: 7255bf8d8aff02f4551a691fb699b8b854db45bd02f73e2dbbb52c481c2b0c14

Relation: equivalent

Source CMElementaryGeometry.lean:661-674: The formal non-infinite Hausdorff measure is equivalent to finite mass here and follows from the four exact arc-length formulas.

Findings: none.

CMElementary.GeometryBridge.four_lower_chord

Source hash: ea53d14df83b28b6ff1d37f67f32312ef47517db52c766fac334806d2ca49d27

Relation: equivalent

Source CMElementaryGeometry.lean:704-718: The analogous lower chord argument retains the strict horizontal inequality and lower interface height.

Findings: none.

CMElementary.GeometryBridge.four_realization

Source hash: 1b8185e0cc3f160a04acc0e5e04323f2a403c0ab76c247907611a7bdaa609ec8

Relation: equivalent

Source CMElementaryGeometry.lean:748-774: All four-arc realization fields are constructed from the prescribed parameters, including both chord interiors and regularity at the endpoint.

Findings: none.

CMElementary.GeometryBridge.four_regularity

Source hash: 6063139ac078a2209fbe7fc6efd1db58eb52039ae2c5d24774d181000e68c1d8

Relation: equivalent

Source CMElementaryGeometry.lean:675-688: Regularity is proved for every real density and every assembly; the bounded, measurable density is integrable on the finite area and frontier measures.

Findings: none.

CMElementary.GeometryBridge.four_upper_chord

Source hash: a1a7708ba2cae5d73698e1a96285a00a773db00178be03734f2731b4ee4f2e11

Relation: equivalent

Source CMElementaryGeometry.lean:689-703: Strict upper chord points lie in the carrier but outside its exact frontier; only the endpoints can remain exposed.

Findings: none.

CMElementary.GeometryBridge.horizontal_section

Source hash: e06f6eef4be5d4e887790cde04018ec9bd793a5f96e506896654d18400a3ceff

Relation: equivalent

Source CMElementaryGeometry.lean:151-175: The three-way section decomposition retains both nonnegativity hypotheses and both outer side cutoffs, including their endpoints.

Findings: none.

CMElementary.GeometryBridge.offset_section

Source hash: 0f3ecd491b72066b337a40630ff772b2174b32379dec94670d56b28857d29ea3

Relation: equivalent

Source CMElementaryGeometry.lean:515-538: All four section hypotheses and both disk-side cutoffs are retained, including a possibly negative offset parameter.

Findings: none.

CMElementary.GeometryBridge.reflected_cap_below

Source hash: 294e02a2d2dbbc8a935f73af2e7892771244bc90c2d4e857fd32d9d4484129b9

Relation: equivalent

Source CMElementaryGeometry.lean:317-324: Reflection changes the upper height inequality into the closed lower inequality for every real parameter pair.

Findings: none.

CMElementary.GeometryBridge.side_width

Source hash: a690f71ca11be015e8cbefc4396612cc139bee2f73d2713555ec3c9c25c58675

Relation: equivalent

Source CMElementaryGeometry.lean:132-139: The assembly hypotheses supply the density and curvature bounds, and the sine-gap definition gives precisely the displayed width.

Findings: none.

CMElementary.GeometryBridge.sine_gap

Source hash: 972dfd8e12f3ba528b4bc8054ce9f622e3d59c0d76ad543257c792688bb4e54e

Relation: equivalent

Source CMElementaryGeometry.lean:128-131: The sine-of-inverse-cosine identity is unconditional for the total functions, and division squared matches the radical definition.

Findings: none.

CMElementary.GeometryBridge.strict_chord_not_arc

Source hash: b0312da63f9829854c31e4257ff6ffb230e77daca0b49854c001774049378eed

Relation: equivalent

Source CMElementaryGeometry.lean:428-441: Strict chord displacement excludes the circle trace using the chord-radius identity; the two chord endpoints are correctly excluded from the premise.

Findings: none.

CMElementary.GeometryBridge.three_bounded

Source hash: 8dd2208ef6c92ad151b2a77cf2af0f518496c61601318bbba67d9e75d7051721

Relation: equivalent

Source CMElementaryGeometry.lean:410-421: The finite union of the rectangle and disk pieces is bounded; no area or perimeter hypothesis is needed.

Findings: none.

CMVElementary.GeometryBridge.three_frontier_finite

Source hash: 4ff646f8685352a1506b99597ef8d06abcc7865827dbb535b135a57304677708

Relation: equivalent

Source CMVElementaryGeometry.lean: 325-344: The proof derives finite canonical mass from the exact Euclidean frontier pieces, not from a modeled perimeter sum.

Findings: none.

CMVElementary.GeometryBridge.three_realization

Source hash: db0addcbf73e2d7c5cb5db55b0994f694f8583f8e05e0428f700aa9cac29e2be

Relation: equivalent

Source CMVElementaryGeometry.lean: 494-514: The assembly is constructed from the density and open curvature bounds; regularity, both integrals, major angles and the disk branch are derived.

Findings: none.

CMVElementary.GeometryBridge.three_regularity

Source hash: 73f3ce51b28ce02d6cc63d5fb204d846cba402ac345eddf320f580aac4f46265

Relation: equivalent

Source CMVElementaryGeometry.lean: 422-427: The proof transports the literal carrier equality and supplies all six regularity fields from existing geometric results.

Findings: none.

CMVElementary.GeometryBridge.three_strict_chord

Source hash: 4464866837d47e9a8c1cbf531c0302898770abce4a983a8a096cdbf3a49d1a2b

Relation: equivalent

Source CMVElementaryGeometry.lean: 450-493: Membership in the carrier and exclusion from every exact frontier component establish interior membership for strict attaching-chord points.

Findings: none.

CMVElementary.GeometryBridge.three_top_circle

Source hash: 2c42afd2f1041c6eb7aaaecf9fbcab707e24f86eb7b6bf6aaa7c27e8dcdadc9c

Relation: equivalent

Source CMVElementaryGeometry.lean: 442-449: The radius-curvature identity and the inner-angle cosine give the stated top-chord circle equation.

Findings: none.

CMElementary.GreatestRootClaims

Source hash: 2882e2312daec98183445737a88569349612dfd73820523f6968960d175c540c

Relation: equivalent

Source CMElementary.lean:125-140: The same witnesses satisfy compactness, greatestness, the right-hand strict barrier and all support-function claims; none is replaced by uniqueness.

Findings: none.

CMElementary.H

Source hash: 7854edb36a536e3f5422a2368eda17806c34bc8a4785388c6c3b36ab5259bb52

Relation: equivalent

Source CMElementary.lean:57-58: The product subtraction agrees, and the area interpretation is separately restricted to nonzero curvature.

Findings: none.

CMElementary.H_ge_J

Source hash: c17279ea5358a898bb176a8ea0bfb9b69861b60846ef9ece0ade666383e410f5

Relation: equivalent

Source CMElementary.lean:502-512: Both interval endpoints are included; the nonpositive specified derivative produces a non-strict comparison in the correct direction.

Findings: none.

CMElementary.H_lower

Source hash: 720b73434d49824a156db89a9d0ba747d35c36753fbd55b86da3e68c83bdd227

Relation: equivalent

Source CMElementary.lean:535-544: The proof explicitly uses the lower-third tangent route and the upper-interval comparison; the author does not misidentify the separate derivative proof as this dependency.

Findings: none.

CMElementary.J

Source hash: 25d540a43a2aa9c7be06cd85d61e6c7b622dc32f471c48ed20eb065861a68717

Relation: equivalent

Source CMElementary.lean:53-54: Both argument evaluations, their coefficients and the constant subtraction agree.

Findings: none.

CMElementary.J_continuous

Source hash: 84b0b93cd0d9eb84e88353584df2d2027343d5022858b007f7419abca4bc0d41

Relation: equivalent

Source CMElementary.lean:456-459: The formal continuity assertion is global and unconditional in the real density, as translated.

Findings: none.

CMElementary.J_lower

Source hash: 7c17f318bc98780bcbcb8604e45210707c09ca380628497b97638e7a07fb4d393

Relation: equivalent

Source CMElementary.lean:513-534: The closed unit interval and non-strict lower bound agree; the proof uses opposite monotonicity on the two sides of one third.

Findings: none.

CMElementary.J_one

Source hash: b4fb08f5f0734c8c331c867147a539407df5ce79eda91cd380170130cdf58655

Relation: equivalent

Source CMElementary.lean:460-463: The value at the right endpoint is exactly the same constant as at zero, with no density restriction.

Findings: none.

CMElementary.K

Source hash: a10d4858e4823efb7ef74e910633b656d8f41250c171a018f71c6c9e83eb6efa

Relation: equivalent

Source CMElementaryGeometry.lean:30-33: The coordinate height bounds, absolute horizontal bound and total square-root formula coincide.

Findings: none.

CMElementary.K3

Source hash: 2a77d29e87870894cbcb97b678ab985c697c0b3bb4aa43807c4d681411e9c085b

Relation: equivalent

Source CMElementaryGeometry.lean:42-47: The shifted center and affine shape substitution agree exactly in the horizontal-section formula.

Findings: none.

CMElementary.MonotoneTangent.endpoint_tangent

Source hash: 88e52ddc8ea9dcc69a9b432eccd375de5c9f6debf8af47520109cf01915bce90

Relation: equivalent

Source CMElementary.lean:313-330: All four hypotheses are preserved, including the degenerate zero-length interval and the absence of a continuity assumption on the specified derivative.

Findings: none.

CMElementary.MonotoneTangent.even_difference_antitone

Source hash: 97bae7778a3a2dbe35cd6cd7377c12ce6215839e12460f95936b813a45c4b542

Relation: equivalent

Source CMElementary.lean:345-360: Evenness is required only on the stated closed interval, and the strict-increase premise is retained in the reflected order argument.

Findings: none.

CMElementary.MonotoneTangent.h_lower_third_of_monotone_tangent

Source hash: 0e2a91072c4780dbfb00df264aa74c10fd7f6ee762c1b356a804608b807e7ace

Relation: equivalent

Source CMElementary.lean:370-390: All five independent premises and the closed lower-third domain are preserved; the proof invokes the tangent lemma rather than a lower-bound record projection.

Findings: none.

CMElementary.MonotoneTangent.jprime_antitone_lower_third

Source hash: 4a69d4e44ae00cd18b8420c38e3a7a4932e9ddfeacaf9bb7f9417aec67061e0

Relation: equivalent

Source CMElementary.lean:361-369: This remains a conditional interface using evenness and strict increase, not a theorem obtained by assuming the desired lower bound.

Findings: none.

CMElementary.MonotoneTangent.reflected_difference_antitone

Source hash: 94ffae68ac5742cce7dc826290e898045b7e92e6507e0ceebc649344ec938685

Relation: equivalent

Source CMElementary.lean:331-344: The formal strict-increase premise is retained; both affine arguments stay in the required closed interval.

Findings: none.

CMElementary.P3

Source hash: cb5785ad3e43cf3da1bc0fe1a18a37ce203b8840e222dbdc84f4f9f0806b00b4

Relation: equivalent

Source CMElementary.lean:45-48: The factor two, denominator, angle term and width correction agree, including the later bottom-segment interpretation.

Findings: none.

CMElementary.P3_continuousAt

Source hash: 7e674cd3499110482a7c44788eb37aa27ecb4b4c5616c3ed50d965f7e0f157c3

Relation: equivalent

Source CMElementary.lean:640-644: The author preserves continuity at every nonzero real parameter, including the scalar right endpoint.

Findings: none.

CMVElementary.P4

Source hash: fa98d4e58766dded99b172ec727c96faed5ce0f690c12b06cd4d7e96d0228209

Relation: equivalent

Source CMVElementary.lean:51-52: The formal numerator is four times the angle combination, divided by curvature, exactly as stated.

Findings: none.

CMVElementary.Q

Source hash: d3e3e508995be11614f5392de567ba9897b0da4cbe3fbd07c5418d5edee16afe

Relation: equivalent

Source CMVElementary.lean:62-64: The support function uses the fixed four-arc target area and the variable three-arc area in the correct order.

Findings: none.

CMVElementary.RegionRegularity

Source hash: f00a35a7ca4c7b622746574abc71bc8314649524dfbea7af49e098990a986e81

Relation: equivalent

Source CMVElementaryGeometry.lean:70-79: All six regularity fields are retained, including actual integrability and finite canonical frontier mass; no bounded-variation theorem is claimed.

Findings: none.

CMVElementary.S

Source hash: eca01a5287b23fda232f56d07e08db5438afa4ad2b1cdad6f24ae23ee6cbb5f0

Relation: equivalent

Source CMVElementary.lean:36-37: The defining subtraction coincides. The derivative cancellation is used only in the interior calculus of the article.

Findings: none.

CMVElementary.S_reflection

Source hash: 26f33ed7c33bedaef432b4c32e750889ffa3f094e21ef2c9ac3dfd7f1097086c

Relation: equivalent

Source CMVElementary.lean:410-413: The inverse-trigonometric reflection identity is stated for all real parameters, matching the total functions used in the proof.

Findings: none.

CMVElementary.SameCurvatureClaims

Source hash: 92c4827d671dfef577f54bd3e2cd8d72eee9b5c2fe740b8108100be082a5abfd

Relation: equivalent

Source CMVElementary.lean:103-113: Both algebraic identities and both non-strict lower bounds are retained, together with strict positivity of their margins.

Findings: none.

CMVElementary.ScalarIdentificationClaims

Source hash: 223a2325121ec324ac1308ae10d552d69116f31febe80ced7adf5e478138df58

Relation: equivalent

Source CMVElementary.lean:141-154: The radical normalization and all six scalar identifications have the same closed-right-endpoint parameter domains.

Findings: none.

CMVElementary.ScalarStatement

Source hash: c257cd1dfa40523e9c40ad9df9512627dd550fab8ba104a9be0654c1e976617

Relation: equivalent

Source CMVElementary.lean:65-73: The universal density and curvature quantifiers precede the existential competitor, with equal area and both strict positive-margin conclusions intact.

Findings: none.

CMVElementary.ThreeArcCalculusClaims

Source hash: 9f403aa6eca1f9679fb0baa7f3fae671458d71319127b51fe69683462617d8e3

Relation: equivalent

Source CMVElementary.lean:114-124: The record distinguishes continuity through the right endpoint from interior differentiability, divergence and the variational identity.

Findings: none.

CMVElementary.ThreeArcRealization

Source hash: dabdc6347c8e954b7cae2391d1c86b0e87c40ca9fe83f8a5de7e2c23a0179ca2

Relation: equivalent

Source CMVElementaryGeometry.lean:97-116: The existential assembly bridge and all branch-sensitive fields are retained, including the entire bottom frontier and the exact disk identity at one half.

Findings: none.

CMVElementary.Z

Source hash: c84d33665cc18e69dec374d2734337ae96f82e06e26b17b735e6b9f2251c24d9

Relation: equivalent

Source CMVElementary.lean:59-61: The closed interval and exact equality level are preserved, with the same three parameters.

Findings: none.

CMVElementary.area_identity

Source hash: b69223db97a18767cba5904b4c3b135f6c23755f91e40636295e797774659382

Relation: equivalent

Source CMVElementary.lean:580-586: The squared-curvature identity has the correct subtraction order and only a nonzero-curvature hypothesis.

Findings: none.

CMVElementary.auxiliary_claims

Source hash: 40439108bc1575c7407f75b4eb147fcee6626bb19b9ed5061fbd985ae03037d5

Relation: equivalent

Source CMVElementary.lean:545-573: The constructor supplies every auxiliary field from the density hypothesis, including the independently established derivative and limit fields.

Findings: none.

CMVElementary.elementary_route

Source hash: a1b39bd2222e649a050d9ab1986edb3f443ebd0f728e84997914f57699a72387

Relation: equivalent

Source CMVElementary.lean:755-765: The root supplies all ingredient records and all greatest-root witnesses from the density hypothesis; none becomes an additional assumption.

Findings: none.

CMVElementary.exists_greatest_level_root

Source hash: 0789ece1798caf97991b6506267400db996fa20c928696c85876071a12271aa2

Relation: equivalent

Source GreatestLevelRoot.lean:12-48: The strict opposite endpoint inequalities yield a greatest interior level point and a strict barrier on the half-open interval to its right; uniqueness is not assumed.

Findings: none.

CMVElementary.gamma

Source hash: 89a6e7b6d34e63c790ca99225f414245b7afb3c33a52b21c3baa8e107a3e1c53

Relation: equivalent

Source CMVElementary.lean:38-40: The negative integer power agrees with the inverse-square notation; positivity remains a separately proved assertion.

Findings: none.

CMElementary.gamma_pos

Source hash: 62668904d84e7cbf785e00bbc726c316b4b51445dcf41d3119a83f33d568f2ff

Relation: equivalent

Source CMElementary.lean:446-455: The proof uses a positive inverse-cosine angle and the sine bound to obtain strict positivity for every density exceeding one.

Findings: none.

CMElementary.geometric_comparison

Source hash: 6a426578c4a240700c25141374531b26d559de5c1a0468aaffc359064d84ae40

Relation: equivalent

Source CMElementaryGeometry.lean:783-798: The scalar witness remains strictly below one, permitting unconditional realization of both carriers; substituting the four integral identities gives the frozen primary conclusion.

Findings: none.

CMElementary.geometric_realization

Source hash: f642078096f16ab8666c13613957258016cb00512ac7e7875b5d3006f18b1646

Relation: equivalent

Source CMElementaryGeometry.lean:775-782: The proof supplies both complete realization families unconditionally on their respective curvature domains, exactly matching the frozen secondary root.

Findings: none.

CMElementary.greatest_root_claims

Source hash: 53fc167a9298d1af24874c5394cc36a6b816243904575d5d9ae34c2802ef1d73

Relation: equivalent

Source CMElementary.lean:692-754: The proof constructs a positive left endpoint and the greatest root, then derives the strict barrier and support comparison for those same witnesses.

Findings: none.

CMElementary.hasDerivAt_B

Source hash: e1d93dd53807589db2db47f2a40f0b76ab73445942336bd1513b60f62d0be20a

Relation: equivalent

Source CMElementary.lean:414-431: Both inverse-trigonometric derivative terms, including the outer density coefficient, and all interior hypotheses agree.

Findings: none.

CMElementary.hasDerivAt_J

Source hash: 6ea60c2da7853b72c8a574b92d5a4f302da93a6ae6fbe80e9bc57d4bda8e18c6

Relation: equivalent

Source CMElementary.lean:464-473: The derivative is the explicit continuous extension on the open unit interval; the affine chain-rule factor is correct.

Findings: none.

CMVElementary.hasDerivAt_S

Source hash: 6a00be6e98e39b229cb8434cdcde52aad95e9edad573a64b0f990aa6ffc1e07b

Relation: equivalent

Source CMVElementary.lean:432-445: The product-rule cancellation yields the stated derivative under the same strict density and interior argument hypotheses.

Findings: none.

CMVElementary.jprime

Source hash: 7436fde598ecb0a0029bf5d63b6fb219ce3a88490ddc40a38e83b449bcad9edd

Relation: equivalent

Source CMVElementary.lean:55-56: The author correctly distinguishes the explicitly defined function from an endpoint derivative of the auxiliary function.

Findings: none.

CMVElementary.jprime_neg

Source hash: ef17824b8194ad2eb67c2e74af7d4d9917476e38bb12159acac0cb8f06705777

Relation: equivalent

Source CMVElementary.lean:493-501: The strict upper interval and negative sign agree with the absolute-value argument comparison.

Findings: none.

CMVElementary.jprime_pos

Source hash: 518f854f89ae00c586e7e8fb6fde430c62ba1dd0993821abffa9e6d56c20870f

Relation: equivalent

Source CMVElementary.lean:479-487: The strict lower-third interval and positive derivative sign agree with comparison of the reflected argument and the original argument.

Findings: none.

CMVElementary.jprime_third

Source hash: d0c7dbb0422c10f59db756962a83f292b34b4533d9c653ccc1f9b40c626313c7

Relation: equivalent

Source CMVElementary.lean:488-492: The zero value at one third is unconditional in density and follows from evenness.

Findings: none.

CMVElementary.same_curvature_claims

Source hash: 3f33553e4d18e3fbda9a031f7ee7ae0a6105fe8e92768737441091a565b9f2b8

Relation: equivalent

Source CMVElementary.lean:587-605: The two identities and both positive quantitative gaps are constructed on the closed-right-endpoint curvature interval.

Findings: none.

CMVElementary.scalar_comparison

Source hash: a640af5855e5df41a3c4e3a32f7162ed7950996079c4770f3205ede3aa37114d

Relation: equivalent

Source CMVElementary.lean:766-779: The proof combines the strict support improvement with the non-strict same-curvature lower bound, yielding the stated strict perimeter margin and exact area equality.

Findings: none.

CMVElementary.scalar_identification_claims

Source hash: dee7445c85542ef1b0ae33e2658325f6df23a74fbf06639fe17dbbbad41e7d8d

Relation: equivalent

Source CMVElementary.lean:606-635: The proof normalizes radicals and rewrites each frozen scalar formula; no geometric identity is assumed in this scalar identification.

Findings: none.

CMVElementary.sigma

Source hash: 98806701b3f8eb5d60ae9b027ae38c5a4e8f3e80159c19646f8418f1e33424f4

Relation: equivalent

Source CMVElementaryGeometry.lean:28-29: The coordinate map reflects only the vertical coordinate, correctly producing reflection in the horizontal axis.

Findings: none.

CMVElementary.strict_support_comparison

Source hash: 84d7d46ea81553126d7454da55589ffbab4f6aa5e17b969665fa757ca0a1b655

Relation: equivalent

Source GreatestLevelRoot.lean:49-79: Both continuity assumptions, the endpoint area equality, interior strict barrier and pointwise variational derivative witnesses are retained; no endpoint derivative is required.

Findings: none.

CMVElementary.support_identity

Source hash: 4e41e8ef0dec9764529d7bbbd6d571c686395fbef0c4920da247dc563a9bed

Relation: equivalent

Source CMVElementary.lean:574-579: The formal algebraic identity needs only nonzero curvature, and the author preserves unrestricted real density.

Findings: none.

CMElementary.three_arc_calculus_claims

Source hash: 66d348c34396a46da7b0e5c806d3140bd9f3e5f226bcb53a0dc55b40173d5d26

Relation: equivalent

Source CMElementary.lean:679-691: The record is fully constructed from continuity, the variational derivative theorem and the divergence theorem under the density hypothesis alone.

Findings: none.

CMElementary.three_arc_eventuallyEq

Source hash: 46dc3f0498de249294a8597ff6c3dc5893312352af2c59075b4bc7135e14271b

Relation: equivalent

Source CMElementary.lean:645-656: Neighborhood equality, rather than a single point equality, is stated for both scalar functions and is what transports derivatives.

Findings: none.

CMElementary.three_arc_variational_hasDerivAt

Source hash: 3ffff307dbffe97d952d3fb947b083b8ae73f0b09e0a5cf42f65cc1bf735c4d5f

Relation: equivalent

Source CMElementary.lean:657-666: Existence of the area derivative and the perimeter derivative identity are both retained on the open unit interval.

Findings: none.

CMElementary.upperCap3

Source hash: a9e700433ec6f55d3afb3184a3b903dc0222e7e85c81667191ac80eb26edc559

Relation: equivalent

Source CMElementaryGeometry.lean:48-53: The disk is intersected with the upper closed half-plane, preserving the major-segment branch.

Findings: none.

CMElementary.upperCap4

Source hash: de3d1bb9a28a3f39df18922596753333140775f67fd4b89756b1e878de59e811

Relation: equivalent

Source CMElementaryGeometry.lean:34-38: The upper half-plane and closed disk inequality have the same center and radius; geometric parameter restrictions are imposed later.

Findings: none.

CapSide

Source hash: 6863ffc025769eb8169e9582a3d673fe180025902f766c2557257f6937cc7a7e

Relation: equivalent

Source `CMVGeometry.lean`: 92-99: The two constructors distinguish precisely the upper and lower half-plane orientations.

Findings: none.

EuclideanPlane

Source hash: dfa2db26da624a2b5333b886cff8c9cd3151218ea8aa3e3cb39c78359b32470e

Relation: equivalent

Source `CMVGeometry.lean`: 34-37: The type uses the second-power product norm, matching the Euclidean rather than maximum norm interpretation.

Findings: none.

FinitePerimeterRegion

Source hash: 2efb7176d17b946809f4cd07b78237ddd03cd4197c405c38d08e97342106a1a0

Relation: equivalent

Source `CMVGeometry.lean`: 74-82: The record contains a measurable carrier and the two integrability obligations, not an independent perimeter value or a general bounded-variation condition.

Findings: none.

FinitePerimeterRegion.weightedArea

Source hash: 3892310b68bd795c4ddff8f1d276f58be0efcb96c1e8c4fad96ec7d73deea20a

Relation: equivalent

Source `CMVGeometry.lean`: 83-85: The accessor evaluates the actual area functional on the stored carrier; no scalar area is stored in the record.

Findings: none.

FourArcAssembly

Source hash: 1ac9185bddaf1e6a0074fb39d1e29417262cc431ccce861d3c65ec2e5216fdb1

Relation: equivalent

Source `CMVGeometry.lean`: 2491-2498: The constructor requires a core and an acute exterior half-angle, but not a common-radius incidence relation; the author preserves that distinction.

Findings: none.

FourArcAssembly.boundaryMeasure

Source hash: a07e8d5b98aab97a737c50d9ee9eae871142e1adb2b58a3132be8c4b6a093cce

Relation: equivalent

Source `CMVGeometry.lean`: 2525-2529: The auxiliary boundary measure sums precisely four arc measures and no chord measures; canonical identification is deferred.

Findings: none.

FourArcAssembly.boundaryTrace

Source hash: d8944f364e38eaa5c7b77ce182d417bec91ac425efc229739d886be00ad658c6

Relation: equivalent

Source CMVGeometry.lean: 2625-2630: The declared trace contains exactly the four exposed arcs; its equality with the frontier is not built into the definition.

Findings: none.

FourArcAssembly.boundaryTrace_subset_frontier_carrier

Source hash: 160385ee99b171abc63b7cc7cb5ade687af8d5249c1ab26a5150be49b8c4654b

Relation: equivalent

Source CMVGeometry.lean: 3244-3251: The reverse inclusion places every exposed arc point, including junctions, on the actual frontier.

Findings: none.

FourArcAssembly.carrier

Source hash: a55616796db22264c4fe11d48e20ab173ecb661c3bdced26752617ea90d1f816

Relation: equivalent

Source CMVGeometry.lean: 2519-2524: The assembly carrier is the actual union of core and both exterior cap carriers, including shared chords.

Findings: none.

FourArcAssembly.frontier_carrier

Source hash: 7da383d241cb3cc30fc9da784ef7200d9bb87d82e25006745acf6f7f7d86a78f

Relation: equivalent

Source CMVGeometry.lean: 3252-3259: Combining both inclusions gives the exact complete-frontier identity used in the perimeter bridge.

Findings: none.

FourArcAssembly.frontier_carrier_subset_boundaryTrace

Source hash: 0d72e7b21e3528564863313aa2366a2e4806e01c00a4772872911d424abf3cc5

Relation: equivalent

Source CMVGeometry.lean: 2795-2819: The inclusion excludes unexposed attaching-chord interiors from the assembled frontier, rather than assuming a four-arc boundary.

Findings: none.

FourArcAssembly.isClosed_carrier

Source hash: 7c8404a543b7f83e631b837fdeb1bbef0f479b4e4161ccfe69e2da8d64a80c0a

Relation: equivalent

Source `CMVGeometry.lean`: 2820-2823: The full assembly carrier is closed as a finite union of the three closed components.

Findings: none.

FourArcAssembly.lowerCap

Source hash: a60a9393b363872f08f12dbab41380b51afefa746b9504c5841f28d389091bfe

Relation: equivalent

Source `CMVGeometry.lean`: 2509-2518: The lower cap uses the reflected midpoint and lower orientation while preserving the upper cap's chord and angle.

Findings: none.

FourArcAssembly.measurableSet_carrier

Source hash: bed9e29097b5f894c003225893e3653e7d99d34980093208388840d2a9198ac1

Relation: equivalent

Source `CMVGeometry.lean`: 2549-2553: Measurability follows from the measurable core and caps and is asserted for every assembly.

Findings: none.

FourArcAssembly.outer_radius_eq_core_radius

Source hash: e5fdf5b4f19a967ab6a1b262eade841eb117d850d615f26e5d93bdf93af65de

Relation: equivalent

Source `CMVGeometry.lean`: 3260-3280: The explicit chord-curvature incidence premise is retained and gives equality of upper cap and core radii by positive-denominator cancellation.

Findings: none.

FourArcAssembly.upperCap

Source hash: 3f084f0dbec6bee8992e556fe2ca304aae5eb8981815636bf15454a3b8e613f9

Relation: equivalent

Source `CMVGeometry.lean`: 2499-2508: The upper cap inherits the positive core chord, acute angle and upper interface midpoint with the correct orientation.

Findings: none.

FourArcAssembly.weightedArea

Source hash: ad43970e4ac79b935baed41701092f4d901053e35bb88583dcbd5a9003db13a3

Relation: equivalent

Source `CMVGeometry.lean`: 2541-2545: The area definition evaluates the ambient weighted integral over the carrier rather than storing its eventual formula.

Findings: none.

FourArcAssembly.weightedArea_formula

Source hash: 839e3c1d7a83b5801ac118ef77feb57b415f9e5a71dfa154b8987ba3b91118a0

Relation: equivalent

Source `CMVGeometry.lean`: 2554-2602: The actual weighted-area formula weights only the exterior caps and holds for every real density, using planar-null overlaps.

Findings: none.

FourArcAssembly.weightedPerimeter

Source hash: df8ef218db7076c4681d4506265ae824cfa4b1a2ec872c320aeb4f283b25493b

Relation: equivalent

Source `CMVGeometry.lean`: 2546-2548: The author explicitly labels this parametrized-measure perimeter as auxiliary, avoiding confusion with its later canonical identification.

Findings: none.

FourArcAssembly.weightedPerimeter_formula

Source hash: 43ff4cddb6774032399c0e7d7f5767a18500986a522e0ace1e470e2035290acd

Relation: equivalent

Source `CMVGeometry.lean`: 2603-2624: The assertion evaluates the auxiliary arc-measure perimeter, weighting both cap lengths and leaving core arcs at density one.

Findings: none.

FourArcCandidate

Source hash: 24dc0ff22e0918bdb2c7f16fab0da52bc08326d88e5cb8e1d2e8a8f7ec2bee62

Relation: equivalent

Source `FourArcCandidate.lean`: 23-35: The constructor bounds on curvature and acute cap angle agree; density jump and Snell incidence are correctly left to a separate predicate.

Findings: none.

FourArcCandidate.SatisfiesCMVTypeIVHypotheses

Source hash: c09b9c7180b6131040333c0b3c2623f3c89af7f54fe8ab0b9f6a41be873ca0dc

Relation: equivalent

Source `FourArcCandidate.lean`: 84-89: The predicate contains exactly density contrast and the physical cosine incidence; both are stated explicitly.

Findings: none.

FourArcCandidate.WeightedArea

Source hash: 8256dd516e32d853aaed6a85d09fdb0b62411bd0bf67f1df7ffd123e47baed99

Relation: equivalent

Source `FourArcCandidate.lean`: 76-78: Unfolding the accessor gives the actual weighted carrier area, matching the author's interpretation.

Findings: none.

FourArcCandidate.WeightedPerimeter

Source hash: d649676f2fd717389a6e9a17a385fe854bc399e17ecfdad24bd8a7bcc9a36521

Relation: equivalent

Source `FourArcCandidate.lean`: 79-83: Unfolding the accessor gives the canonical complete-frontier integral, not the auxiliary assembly measure definition.

Findings: none.

FourArcCandidate.assembly

Source hash: ca67fddb2d167afab63544438387902bb645349ebfde9d83c7a2719a2277c941

Relation: equivalent

Source `FourArcCandidate.lean`: 66-72: The assembly uses precisely the constructed core and candidate exterior angle, giving congruent upper and lower caps.

Findings: none.

FourArcCandidate.beta

Source hash: f3520601e601026585d5024a59c3bee3be514d265897b65e66919a9d3ea7fd1e

Relation: equivalent

Source `FourArcCandidate.lean`: 39-41: The complementary angle definition agrees and is distinguished from the core's arcsine side angle.

Findings: none.

FourArcCandidate.candidate_area_formula

Source hash: b6d944ecc5d4bab77cadf263520f194142a9d6855088a9b98e5873be144ab8cc

Relation: equivalent

Source `FourArcCandidate.lean`: 410-430: The area proof uses the Snell incidence to simplify the weighted cap term and the complementary-angle identity, with the exact coefficients stated.

Findings: none.

FourArcCandidate.candidate_perimeter_formula

Source hash: b4949bdc7019f429151f2faf734db8466461943a33be006f4c104caebb493fe0

Relation: equivalent

Source `FourArcCandidate.lean`: 391-409: The physical predicate remains in the theorem's hypothesis even though the arithmetic proof does not use it; the author explicitly retains that contract.

Findings: none.

FourArcCandidate.capArea

Source hash: eb6c25ab8d5e1e4d2ce589b98fc3af2d04e58bd682ad273d1b07834426ee23d0

Relation: equivalent

Source `FourArcCandidate.lean`: 45-49: The scalar cap area is the circular-segment expression divided by squared curvature, with the correct sine-cosine subtraction.

Findings: none.

FourArcCandidate.capArea_eq_chord_sq_mul_area

Source hash: 0e9545ae1662d69d55500dc9264e42c4c5b03b78571f6473a85822aca53c6073

Relation: equivalent

Source `FourArcCandidate.lean`: 104-109: The proof clears positive sine and curvature denominators to identify the two scalar cap-area normalizations.

Findings: none.

FourArcCandidate.capChord

Source hash: c09aa141e8bbabbe7279bce2473dd47aa931f875442fe1fc79d759685536fb1e

Relation: equivalent

Source `FourArcCandidate.lean`: 42-44: The candidate chord formula has the factor two and positive curvature denominator needed for the common radius.

Findings: none.

FourArcCandidate.four_arcs_common_radius

Source hash: cb52732d26d7e21d171f55b06767618effe21e04fd18d38de3625ce3799422be

Relation: equivalent

Source `FourArcCandidate.lean`: 167-181: Both exterior radii equal the core radius from the chord construction alone; the author correctly avoids adding a density or Snell assumption.

Findings: none.

FourArcCandidate.lowerCap_euclideanArea

Source hash: 14a860cf99daaf759c3fa7838aed86edfb2879de1a5a3f6f7d84f9996c7da155

Relation: equivalent

Source `FourArcCandidate.lean`: 159-166: The reflected lower cap has the same chord-angle area, and the orientation change introduces no sign in area.

Findings: none.

FourArcCandidate.region

Source hash: 8ab9f84c197a1a6dbe90236469e9ff5cc2696f04e42133db44cdf34546558b75

Relation: equivalent

Source `FourArcCandidate.lean`: 73-75: The candidate is embedded in the four-arc constructor of the competitor datatype without adding a minimality claim.

Findings: none.

FourArcCandidate.side_caps_area_formula

Source hash: a639bd8c9eef40f2c4f87fdb62b671cf7db0ce41013e7d923721f6f472bc8cf1

Relation: equivalent

Source `FourArcCandidate.lean`: 351-363: The normalized side-cap area simplifies using sine and cosine of principal arcsine, including the unit-curvature endpoint.

Findings: none.

FourArcCandidate.sin_alpha_pos

Source hash: b92e14d3743fcaa1d6c447d96c6334ba3f5f129152aa26ff47b97d6fd8de2e28

Relation: equivalent

Source `FourArcCandidate.lean`: 95-98: The acute positive exterior angle gives the strict sine positivity with no additional density premise.

Findings: none.

FourArcCandidate.stripCore

Source hash: 66aff9e1eb934bbbeeb8dd09a1a68a4c23aef3ee09735d3ba9bc12eadb4664ee

Relation: equivalent

Source `FourArcCandidate.lean`: 53-65: The core construction uses the candidate chord and curvature and proves the required constructor bounds, retaining unit curvature.

Findings: none.

FourArcCandidate.upperCap_euclideanArea

Source hash: 63eb34459de3e09e8c932883edd9446c50efc2cc1a148ef60c5c77cbd56d7496

Relation: equivalent

Source `FourArcCandidate.lean`: 153-158: The upper cap's chord and angle are those of the candidate, so the normalized scalar area becomes the candidate cap area.

Findings: none.

FourArcCandidate.weightedArea_components

Source hash: 654554fba4e2f4847eae4f853550061afa29cf158da02de52109cffffacedfd702e

Relation: equivalent

Source `FourArcCandidate.lean`: 364-373: The actual assembly area formula and equality of the two cap areas yield the stated weighted component sum.

Findings: none.

FourArcCandidate.weightedPerimeter_components

Source hash: d02953cec6a51d470b1b940fa2318bbd5c86354d0e420d179322c86356c9434a

Relation: equivalent

Source `FourArcCandidate.lean`: 374-390: The proof first invokes the canonical frontier bridge and then the auxiliary component formula, preserving the perimeter's geometric meaning.

Findings: none.

FrontierMeasure

Source hash: 7323bb80c0a565daf410d27d199e2143ee33a0ecca0460c52504f2e7e50b48e2

Relation: equivalent

Source `CMVGeometry.lean`: 61-66: The measure is determined by Euclidean Hausdorff measure restricted to the complete frontier and transported back; it is not selectable boundary data.

Findings: none.

HorizontalSegment

Source hash: e7bebc91d2e392d80b21e5e8b5487591a5477fe10be16627e5b6180dd120506

Relation: equivalent

Source `CMVGeometry.lean`: 392-399: The segment primitive requires strictly positive length; the explanation correctly separates it from the possibly degenerate bottom segment.

Findings: none.

HorizontalSegment.carrier

Source hash: 094b96b3006204081e6d336250f56e63f6b337cc71255210b9bae8ae3b894041

Relation: equivalent

Source `CMVGeometry.lean`: 406-408: The absolute displacement bound describes the closed horizontal segment with both endpoints included.

Findings: none.

HorizontalSegment.euclideanLength

Source hash: c965c7e2ff7089171e9318c88e3733eb9629da0051af6a9aad082bf1ad5a5aef

Relation: equivalent

Source `CMVGeometry.lean`: 409-415: The scalar length is definitionally the positive chord parameter; geometric Hausdorff length is proved later.

Findings: none.

HorizontalSegment.leftEndpoint

Source hash: d64f9e0ea48cfbbc1b06bed874068659f0011957eca2b8226c738430113a0615

Relation: equivalent

Source `CMVGeometry.lean`: 400-402: The left endpoint is the midpoint minus half the positive chord length in the horizontal coordinate.

Findings: none.

HorizontalSegment.rightEndpoint

Source hash: a14791633639531ab0cb8801c8e0e16383dd7712b7d5730aa7f0d9178c83e5da

Relation: equivalent

Source `CMVGeometry.lean`: 403-405: The right endpoint is the midpoint plus half the positive chord length, with the base height unchanged.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeAngle

Source hash: 6ec918819525fbc766cbbf40969985e480000f0bcbcf23d536b033e9e80b6380

Relation: equivalent

Source `LeanSuffixAnalytic.lean`: 154-205: Both radical denominators are positive under the same open curvature and density hypotheses; the chain-rule factor two is retained.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeDelta

Source hash: 6a259e36542b424c2657b96b59d66f9d38d054281073fe5c590c3110a33522c8

Relation: equivalent

Source `LeanSuffixAnalytic.lean`: 206-240: The derivative coefficient includes both the affine factor and the density in the outer radical denominator, with no endpoint derivative assertion.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeShape

Source hash: fc5b59d714869fde00b69f1950c83bd7a32c47524e01ad8d04455ff346c39d58

Relation: equivalent

Source `LeanSuffixAnalytic.lean`: 146-153: The affine shape derivative exists at every real parameter and equals the stated constant.

Findings: none.

LeanSuffixAnalytic.typeFourAngle

Source hash: 94e6a9f5b491cb7ca4802d89fba1fa7ccb864bbe777d6e41d89b1d6bf892bdd0

Relation: equivalent

Source `LeanSuffixAnalytic.lean`: 53-56: The four-arc angle formula is exactly the article's angle combination at the curvature parameter.

Findings: none.

LeanSuffixAnalytic.typeFourArea

Source hash: 775f7921b66716563dd5e0e4c35562d880f5cef2a71b40cef616cd10eb9488d2

Relation: equivalent

Source LeanSuffixAnalytic.lean:61-64: The normalized area retains the factor two and curvature-weighted radical correction over squared curvature.

Findings: none.

LeanSuffixAnalytic.typeFourDelta

Source hash: a4c8b30971f59f8f04626dbddf49af92c98dbfa9126bf3766c0be8b6472cbfb3

Relation: equivalent

Source LeanSuffixAnalytic.lean:57-60: The first radical is normalized outside the square root; its positive-density identification with the article's function is stated separately.

Findings: none.

LeanSuffixAnalytic.typeFourPerimeter

Source hash: ad0bebbd79c17351c0f12ce6a7fdae3e930dba33de0077415a6116df1d1e6e25

Relation: equivalent

Source LeanSuffixAnalytic.lean:65-70: The normalized four-arc perimeter is definitionally the article's formula, requiring no radical identification.

Findings: none.

LeanSuffixAnalytic.typeThreeAngle

Source hash: fb7617b8b745792c6f019f3cbcf3c26d5e54f1fc7dd47bb69cec919c712d110a

Relation: equivalent

Source LeanSuffixAnalytic.lean:35-38: The normalized three-arc angle is the article's angle combination evaluated at the affine shape parameter, with identical density scaling.

Findings: none.

LeanSuffixAnalytic.typeThreeArea

Source hash: 63d13d5879252e16ac6f53a2d4449b4f511e353d6016e98158793484dcfaa8f7

Relation: equivalent

Source LeanSuffixAnalytic.lean:44-48: The coefficient of the radical correction is the shape parameter plus two, agreeing with the article after the affine substitution.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_tendsto_atRight_zero

Source hash: e82ffa97f750ea7c64d3d22eebba58e03fe2fc0f731297b9787ae711ded5a49a

Relation: equivalent

Source LeanSuffixAnalytic.lean:828-871: The numerator has the stated positive limit, and the proof compares against a diverging positive reciprocal on the right of zero.

Findings: none.

LeanSuffixAnalytic.typeThreeDelta

Source hash: f96722ed089f4e5911c6013a0aa601eb4080090036491702cd0fa78e3e31113f

Relation: equivalent

Source LeanSuffixAnalytic.lean:39-43: The radical normalization includes division by density outside the first square root; the author correctly conditions its identification with the article's radical difference on positive density.

Findings: none.

LeanSuffixAnalytic.typeThreePerimeter

Source hash: 22e35e36f9c0930c9f9233f0004c686c48f6f282298a578def5a6e6bba11f34e

Relation: equivalent

Source LeanSuffixAnalytic.lean:49-52: The normalized perimeter preserves the factor two, angle constant and radical correction, including the bottom-segment term after geometric realization.

Findings: none.

LeanSuffixAnalytic.typeThreeShape

Source hash: 9ef91461acc972465b98decfe31d601788f1354eb0ffbb8d094be6f5c46ca593

Relation: equivalent

Source LeanSuffixAnalytic.lean:32-34: The affine shape parameter and its total real domain agree exactly.

Findings: none.

LeanSuffixAnalytic.typeThree_variational_hasDerivAt

Source hash: 727f5f60431aed129e76617edde1e3ccdf8d18a1ae71fbf80ca33147d09b9cfa

Relation: equivalent

Source LeanSuffixAnalytic.lean:554-639: The proof differentiates the explicit quotients and uses the radical cancellation to establish both derivative existence and the variational identity, not global area monotonicity.

Findings: none.

OneSidedCircularCap

Source hash: a9dd2ba7ee653be367b6a2072de1384c3362b0214117e3aecceae763bc6f8403

Relation: equivalent

Source CMVGeometry.lean:100-112: The positive chord and full open half-angle interval are preserved, so major caps are not excluded.

Findings: none.

OneSidedCircularCap.arcLength

Source hash: 60daa6fa20eb20410aecf4ceebd5a41e50c48a0eb958e6b55f1bcb20cd85ff25

Relation: equivalent

Source `CMVGeometry.lean`: 167-170: The defined scalar measures only the selected arc, not the chord, and agrees with the radius-angle expression under admissibility.

Findings: none.

OneSidedCircularCap.arcLength_pos

Source hash: 615464c0ac73967d872be72b9e5d931fb0644d2db70b6e9168ad1dd7c81aa184

Relation: equivalent

Source `CMVGeometry.lean`: 220-222: The selected arc length is positive because both factors in its definition are positive.

Findings: none.

OneSidedCircularCap.arcPoint

Source hash: 2f12b44987266410d1b21333c27225fca467817287b4028245b465856f5540c3

Relation: equivalent

Source `CMVGeometry.lean`: 124-133: Both coordinate branches and their total parameter domain agree; restriction to the selected arc is described separately.

Findings: none.

OneSidedCircularCap.arcPoint_in_closed_side

Source hash: 4692c99be4beede3a58f1e164acd1cfe21db7a9160da63340de9490acd46c184

Relation: equivalent

Source `CMVGeometry.lean`: 240-252: The closed parameter bound yields the correct non-strict side inequality for both orientations, including endpoints.

Findings: none.

OneSidedCircularCap.arcPoint_in_open_side

Source hash: 3dbf6a865d1d6331f9c3406c05af249acf00d7c487349d982b588394a814d697

Relation: equivalent

Source `CMVGeometry.lean`: 253-266: The strict parameter bound yields the strict side inequality for both orientations on the whole cap-angle domain.

Findings: none.

OneSidedCircularCap.arcPoint_left

Source hash: 4e22ae8901c6f895322d9f1591e370b613671c823882adf707a5fac1ae750287

Relation: equivalent

Source `CMVGeometry.lean`: 228-232: The negative-angle parameter gives the left endpoint in either orientation by odd sine and even cosine.

Findings: none.

OneSidedCircularCap.arcPoint_right

Source hash: ba639553cc75debbcac47cb7f164b0cdee0e14e9586cdc6d1cf857f7fa27f21b

Relation: equivalent

Source `CMVGeometry.lean`: 233-239: The positive-angle parameter gives the right endpoint in either orientation, with zero vertical displacement from the base.

Findings: none.

OneSidedCircularCap.arcTrace

Source hash: 5e3edb81f4f16ef8dbff7c15d5756a176b1f99c3bd5fb86ce703134a0a727a89

Relation: equivalent

Source `CMVGeometry.lean`: 656-661: The trace replaces the disk inequality with circle equality while retaining the closed side condition.

Findings: none.

OneSidedCircularCap.carrier

Source hash: e0644dd0b5f330958b79efbb3810eb1cc2f4356e5db251a95736b532b75bfcf1

Relation: equivalent

Source `CMVGeometry.lean`: 146-152: The closed disk and orientation-dependent closed half-plane conditions are both retained, including the chord.

Findings: none.

OneSidedCircularCap.center

Source hash: 58a018a3af71f1eb6c5aebb7ee395b60aab79f7b55a9d384d43694baace0d346

Relation: equivalent

Source `CMVGeometry.lean`: 117-123: The center signs switch with orientation and retain the cosine sign change for major caps.

Findings: none.

OneSidedCircularCap.chordCarrier

Source hash: 0705736836c9203b120221cb0770c365bada15ab749e72d0791d55f58eba4c1e

Relation: equivalent

Source `CMVGeometry.lean`: 160-163: The closed chord includes both endpoints and has the same midpoint and half-length as the formal set.

Findings: none.

OneSidedCircularCap.chordCarrier_subset_carrier

Source hash: a35dcecd8fd218da748131deaec9b02f5328c1034952ad269b6ba1f5b686d947

Relation: equivalent

Source `CMVGeometry.lean`: 311-334: Every point of the closed chord satisfies the disk and half-plane conditions, including both endpoints.

Findings: none.

OneSidedCircularCap.ell_theta_pos

Source hash: 88c4230a8ce8dbe3ed5e64385c18de0588039e6c242ead1e2df6299d087f0ad8

Relation: equivalent

Source `CMVGeometry.lean`: 216-219: The positive ratio uses both positive half-angle and sine; no endpoint assertion is introduced.

Findings: none.

OneSidedCircularCap.euclideanArea

Source hash: 5c476b88f35c9e9f6afc55ed7ed9ef6be1395a7b5a3003452ff8691443c49da

Relation: equivalent

Source `CMVGeometry.lean`: 164-166: The definition is a scalar normalized area; equality with carrier volume is correctly deferred to the volume theorem.

Findings: none.

OneSidedCircularCap.euclideanArea_pos

Source hash: 3902ccdc9eb3694014c5a47aad3179ebe09a3fd3cddb066b9c4c1ca933463218

Relation: equivalent

Source `CMVGeometry.lean`: 213-215: The statement concerns scalar Euclidean area, obtained from positive chord square and normalized segment area.

Findings: none.

OneSidedCircularCap.frontier_carrier

Source hash: d9cb2c749721b51457faae17a95033cf908527d3fdc652ebedacfb2292d579d

Relation: equivalent

Source `CMVGeometry.lean`: 697-761: The exact frontier includes both the selected circle arc and the full base chord; the author does not omit the chord at this primitive level.

Findings: none.

OneSidedCircularCap.interiorCarrier

Source hash: d9986ab2dd520dfd41d6ccb99ca9878df085185d6c10c136685b43549b4c4dc1

Relation: equivalent

Source `CMVGeometry.lean`: 153-159: Both inequalities are strict; the author distinguishes this auxiliary open carrier from the separately proved topological interior inclusion.

Findings: none.

OneSidedCircularCap.interiorCarrier_subset_interior

Source hash: e055c7441e3e2230b6bd3c3989fc3043eb631862e9cdc4e7fe3a7963b0242ddc

Relation: equivalent

Source `CMVGeometry.lean`: 687-696: The assertion is an inclusion into the topological interior, not an unsupported equality of the two interior descriptions.

Findings: none.

OneSidedCircularCap.isClosed_carrier

Source hash: d757ff8cbf988a97323a1f1c27dc667eab9b6475f2eb7d1c16beffa82e3009a4

Relation: equivalent

Source `CMVGeometry.lean`: 364-391: The closed disk and closed half-plane conditions give a closed carrier for either orientation.

Findings: none.

OneSidedCircularCap.isOpen_interiorCarrier

Source hash: ec8c11c26c05148d770698c167cdec56869de41dc90864a365cf2514de87bc1

Relation: equivalent

Source `CMVGeometry.lean`: 662-686: Strict inequalities of continuous coordinate functions give an open set in either cap orientation.

Findings: none.

OneSidedCircularCap.leftEndpoint

Source hash: a193f337e9021d7e081f4666c9f784092558b2268ec0e9060f11b3d18dfc3c8d

Relation: equivalent

Source `CMVGeometry.lean`: 134-137: The endpoint uses the negative half-chord displacement on the base line, exactly as defined.

Findings: none.

OneSidedCircularCap.measurableSet_carrier

Source hash: e0f551f525fbf0409babb736205ec484e0a0c2ff7039ce55de1e7a7fc5595a27

Relation: equivalent

Source `CMVGeometry.lean`: 335-363: Measurability is asserted for every admissible coordinate cap, not supplied as extra construction data.

Findings: none.

OneSidedCircularCap.mem_chordCarrier_of_mem_carrier_of_eq_base

Source hash: 424261710f758c71ebe7e57dd13ebd11ffc59cb51b840ff8fc86319890ddae19

Relation: equivalent

Source `CMVGeometry.lean`: 291-310: The disk section at the base line is bounded by the closed half-chord; the author retains carrier membership as a premise.

Findings: none.

OneSidedCircularCap.radius

Source hash: e898cc7066a75a606fdc3b1d1e563939ad569bdb0f66fd289569e58c7f4c44b8

Relation: equivalent

Source `CMVGeometry.lean`: 113-116: The radius formula has the correct factor two and sine denominator, whose positivity follows from cap admissibility.

Findings: none.

OneSidedCircularCap.radiusSquaredAt

Source hash: a1c0f970b426c7f0e6a013d8d8bbb613fbb3b5c253445dacecdb9b47cb6785a

Relation: equivalent

Source `CMVGeometry.lean`: 142-145: The expression is the sum of squared coordinate differences, so its interpretation is Euclidean even in the coordinate product type.

Findings: none.

OneSidedCircularCap.radius_mul_sin

Source hash: 4b800b6287f8bc78c34b40f39fff02ab7d25631ad2fd22f8819095ceda60143e

Relation: equivalent

Source `CMVGeometry.lean`: 223-227: Cancellation of the nonzero sine yields the exact half-chord identity with the correct factor two.

Findings: none.

OneSidedCircularCap.radius_pos

Source hash: 7bcdcf3411946f45cb3c854e76728bf4e1477fc16c0bd8673b5d0f6f904d6adf

Relation: equivalent

Source `CMVGeometry.lean`: 210-212: Positive chord and positive sine denominator give a strictly positive radius, as stated.

Findings: none.

OneSidedCircularCap.rightEndpoint

Source hash: e83d4942af5a95dd72bd914bc15f416e046e239fc23cafff7a851807f832f26a3

Relation: equivalent

Source `CMVGeometry.lean`: 138-141: The endpoint uses the positive half-chord displacement on the base line, exactly as defined.

Findings: none.

OneSidedCircularCap.sin_theta_pos

Source hash: 8c22795350624813eb7b9aec2fdcda7bfc438d144ebb38b607db3cac77688583

Relation: equivalent

Source `CMVGeometry.lean`: 207-209: The strict sine positivity uses precisely the cap's open half-angle interval.

Findings: none.

OneSidedCircularCap.volume_carrier

Source hash: 0d51f9d06b45af9eac50b2062d616aee51a260200ad5a2e96d2ecbe8e98be85c

Relation: equivalent

Source `CMVGeometry.lean`: 1996-2028: The actual cap carrier volume is identified with the scalar normalized area in the extended nonnegative reals.

Findings: none.

PlanePoint

Source hash: b6c33ab7d9e6652b4833188977c0694ee665f31ebd07e572ea7a67ee7a8a0183

Relation: equivalent

Source `CMVGeometry.lean`: 30-33: The coordinate type is the real product, and the explanation correctly distinguishes its default metric from the Euclidean perimeter metric.

Findings: none.

ReplacementAssembly

Source hash: e0b73790946058b82b8d6a4bdde47ad7560092e26acf3d58cb0c91dc0cfa445f

Relation: equivalent

Source `CMVGeometry.lean`: 3281-3288: The full open replacement-angle range is retained; this carrier datatype is not misrepresented as the final comparison argument.

Findings: none.

ReplacementAssembly.carrier

Source hash: 36b80491dc0fecbf358d87c6bfe7194faf4bcad4ac91372874e8af60a30fd23e

Relation: equivalent

Source `CMVGeometry.lean`: 3305-3309: The definition is only the union of the core and upper cap; no greatest-root or optimality claim is attached to it.

Findings: none.

ReplacementAssembly.upperCap

Source hash: b30d86506333a29d0fea0a0c213d5ab5e404eb8902dd90c6019d57dad3b8f56f

Relation: equivalent

Source `CMVGeometry.lean`: 3289-3298: The replacement upper cap uses the core chord and upper interface, with the full cap-angle admissibility conditions.

Findings: none.

ReplacementAssembly.weightedArea

Source hash: 30517c93164485b2476e45ea547b19187ad53db44a33af2bf70144ca4c93c662

Relation: equivalent

Source `CMVGeometry.lean`: 3326-3329: The replacement area's definition is the actual carrier integral, not a stored scalar comparison witness.

Findings: none.

StripCore

Source hash: a32ba501747725a0a1307cf7bd2abbffd387c8bc9d810f24cf348c7c04513aad

Relation: equivalent

Source `CMVGeometry.lean`: 416-425: The core data retain positive chord and curvature with a non-strict upper curvature bound.

Findings: none.

StripCore.boundaryArcLength

Source hash: 74e634700740d7651d4ed773b1cb3cb71b11c7eedacd0c71b6a84a0e6ec43665

Relation: equivalent

Source `CMVGeometry.lean`: 457-459: The scalar boundary length includes the two exposed side arcs and excludes both attaching chords.

Findings: none.

StripCore.carrier

Source hash: 05f714bedd5ca3f09fd1dac168964dc190e4830b8099d994e590458d8167e259

Relation: equivalent

Source `CMVGeometry.lean`: 449-452: The formal carrier is the union of three coordinate pieces, not an area-defined abstract region.

Findings: none.

StripCore.carrier_y_bounds

Source hash: dfd3510aaac02f0e3230984bb1d2516c15f954b7d416fa7e30c71ad44efa3ab8

Relation: equivalent

Source `CMVGeometry.lean`: 482-490: All three carrier components explicitly impose the closed strip-height condition.

Findings: none.

StripCore.cos_sideAngle

Source hash: 5c26f39a6dccf790fede705362af68051955e479d2108518ff90a5d9985148ed

Relation: equivalent

Source `CMVGeometry.lean`: 478-481: The principal cosine is the nonnegative square root of the stated radicand.

Findings: none.

StripCore.euclideanArea

Source hash: d05ac0be663f49bfb5425917ab3d2dcdd3367e8ef25ec467ccc3d1ef4ccf07a7

Relation: equivalent

Source `CMVGeometry.lean`: 453-456: The scalar area adds the rectangle and two normalized chord-two cap areas, with the correct coefficient eight.

Findings: none.

StripCore.frontier_carrier

Source hash: 34bdf246812f32ba3d6167bf3b3ff5fea490b0816170740d97a7d77e703585bf

Relation: equivalent

Source `CMVGeometry.lean`: 1088-1199: The exact frontier consists of the two exposed arcs and both horizontal chords; internal rectangle-side seams are not counted.

Findings: none.

StripCore.isClosed_carrier

Source hash: d28fca8c993d5d2b9aa0b2ebe7f3f0372ebdd951a5cda90d26adf748bface0e4

Relation: equivalent

Source `CMVGeometry.lean`: 805-843: The coordinate core is a finite union of closed pieces, so the stated closedness has no extra regularity premise.

Findings: none.

StripCore.leftArcTrace

Source hash: 702872b40533619867ab566579d52d3f6a44e812910796d96106d1c50acbbfc2

Relation: equivalent

Source `CMVGeometry.lean`: 762-766: The circle equality and left cutoff select the exposed left trace with the closed height range.

Findings: none.

StripCore.leftCapCarrier

Source hash: 2ca301cf56be587995c60fad814a6021c666b20d0f45e1fa3f2964b23f678cbc

Relation: equivalent

Source `CMVGeometry.lean`: 440-443: The left disk inequality, left exterior cutoff and strip-height condition are all present.

Findings: none.

StripCore.leftCenterX

Source hash: 01fd3b14ca7e8a9d341e71e2fb86b773a9bea051016f7f6499489c3b65bab474

Relation: equivalent

Source `CMVGeometry.lean`: 431-433: The left center has the correct sign of the half-chord and cosine displacement.

Findings: none.

StripCore.lowerChordTrace

Source hash: faa702d9afdf2eed044fad4474692be94c38bb3a9868779993853eb74445596

Relation: equivalent

Source `CMVGeometry.lean`: 774-776: The lower chord trace is the full closed chord on the lower interface, before assembly.

Findings: none.

StripCore.measurableSet_carrier

Source hash: c4a110b00efc328d9afc36bfb523f064a5cb69170025121ec86c0022fc798171

Relation: equivalent

Source `CMVGeometry.lean`: 500-551: Measurability of the finite union follows from the coordinate inequality components; no new assumption is introduced.

Findings: none.

StripCore.radius

Source hash: f23d45618cc8f0a402155f6b0b9c9bed86df54007d8b97c4f4027753ebfcd373

Relation: equivalent

Source `CMVGeometry.lean`: 426-428: The reciprocal radius definition agrees and includes the unit-radius endpoint permitted by the constructor.

Findings: none.

StripCore.radius_pos

Source hash: 754f25da7343b067aca1ccf07fea956e09bfd461a16a072435b6b938ea98f462

Relation: equivalent

Source `CMVGeometry.lean`: 460-462: The strictly positive curvature gives a strictly positive reciprocal radius.

Findings: none.

StripCore.rectangleCarrier

Source hash: 5204ba429abcba2c9a7bfd26b44272d1068d5b57b8a97a88331831661e9408f9

Relation: equivalent

Source `CMVGeometry.lean`: 437-439: The rectangle has the closed chord-width interval and closed strip-height bound, giving the stated area.

Findings: none.

StripCore.rightArcTrace

Source hash: 059982ba84229cdd58242cd7f98720be73ad274e207529f914e522d73ba5cbeb

Relation: equivalent

Source `CMVGeometry.lean`: 767-770: The circle equality and right cutoff select the exposed right trace with the closed height range.

Findings: none.

StripCore.rightCapCarrier

Source hash: 0f7231f5e97656f30e8e5131e2afc098206a3ff9953e2cd02ae8207ac4d54b12

Relation: equivalent

Source `CMVGeometry.lean`: 444-448: The right disk inequality, right exterior cutoff and strip-height condition are all present.

Findings: none.

StripCore.rightCenterX

Source hash: ff1acc9e9b3ef170758d23460b0f614f6eb1c034ebd024b460f189e9f303f923

Relation: equivalent

Source `CMVGeometry.lean`: 434-436: The right center is the horizontal reflection of the left center, with the signs reversed exactly.

Findings: none.

StripCore.sideAngle

Source hash: 4f6b6599037149db7ce42f881c45d0eb7f35fbc43f124df7507d6a83a6d6e15b

Relation: equivalent

Source `CMVGeometry.lean`: 429-430: The principal arcsine defines the side half-angle, with the correct closed upper angle endpoint.

Findings: none.

StripCore.sideAngle_le_pi_div_two

Source hash: 0c0857ab96c23693b4448efce2e70a6069dda6f460cf0118b7e365f0f1074987

Relation: equivalent

Source `CMVGeometry.lean`: 466-469: The formal angle bound is non-strict, retaining the semicircular side arcs at unit curvature.

Findings: none.

StripCore.sideAngle_lt_pi

Source hash: 4d165924e20daf34cd0cd3a93af12e6637e356347b723990beb9d1a85af59a9e

Relation: equivalent

Source `CMVGeometry.lean`: 470-473: The upper arcsine bound is strictly below a full half-turn, as required by the general cap infrastructure.

Findings: none.

StripCore.sideAngle_pos

Source hash: 20886da5ba412dbc0666239218b14d32a1bad68cf7a0f4837b3f6216a46aefbd

Relation: equivalent

Source CMVGeometry.lean: 463-465: The principal arcsine of the positive admissible curvature is strictly positive.

Findings: none.

StripCore.sin_sideAngle

Source hash: e85f9cb3c0c2b22c1ce74fdc60660033205e7a0821e76cfdd7248571352ea184

Relation: equivalent

Source CMVGeometry.lean: 474-477: The sine-of-arcsine identity uses the admissible curvature interval and gives the stated curvature exactly.

Findings: none.

StripCore.upperChordTrace

Source hash: 71db3f396f38ac0c4ed32b75273f08b0092359d4c0fdd7bc77976000e7791fb0

Relation: equivalent

Source CMVGeometry.lean: 771-773: The upper chord trace is the full closed chord on the upper interface, before assembly.

Findings: none.

StripCore.volume_carrier

Source hash: a426f0908aec7b73455f4a0093170232e841cf46ab8f580ee2fe442ff66a7997

Relation: equivalent

Source CMVGeometry.lean: 2202-2242: The rectangle and two side-cap volumes add after null line overlaps, giving the exact scalar core area.

Findings: none.

StripCore.volume_leftCapCarrier

Source hash: ee7dbe308fa5b9feabca0b7d1f31d2c91cdf7e530a43dcc64a52dce7563fb7b8

Relation: equivalent

Source CMVGeometry.lean: 2076-2138: The left side cap has chord two, so the actual planar volume is four times the normalized cap area.

Findings: none.

StripCore.volume_rectangleCarrier

Source hash: b2c1d8716422a2e0169b6aaca489790c14b0594bc75f1e77b6bbec1349d5110

Relation: equivalent

Source `CMVGeometry.lean`: 2060-2075: The rectangle's product interval lengths give twice the core chord, with the correct extended-real embedding.

Findings: none.

StripCore.volume_rightCapCarrier

Source hash: 7bc214af5d8d445665e416853a1fab80112ca5fe1116500de0e9b2b6285dfdf7

Relation: equivalent

Source `CMVGeometry.lean`: 2139-2201: The reflected right side cap has the same actual planar volume and coefficient as the left cap.

Findings: none.

StripCore.weightedArea_formula

Source hash: 22039501f3884648956188fd011ab629e4d948f733db9c19194b8b310134e20a

Relation: equivalent

Source `CMVGeometry.lean`: 2315-2333: The actual weighted integral is the core's Euclidean area because the entire carrier is in the closed density-one strip.

Findings: none.

StripDensity

Source hash: 51e629a8115629e4652f417b9dcfc46beea6a584d8ebcb351e746df181f97ab1

Relation: equivalent

Source `CMVGeometry.lean`: 43-54: The piecewise definition includes both interfaces in the density-one branch and permits arbitrary real density parameters.

Findings: none.

TypeThreeAssembly

Source hash: ba80fb031079587a301c08eeb69ef793b83ab9e2bcde30a061c829264b7e1959

Relation: equivalent

Source `TypeThreeAssembly.lean`: 29-35: The assembly contains precisely the density jump and strict open curvature bounds; all other geometric quantities are determined by those data.

Findings: none.

TypeThreeAssembly.WeightedArea

Source hash: 5d054747c494e793da416a19e2d895420eb9e121366a5645e79b7ab53a0c3c7f

Relation: equivalent

Source `TypeThreeAssembly.lean`: 265-268: The assembly area is defined by the actual ambient weighted integral over its carrier, independently of the modeled scalar expression.

Findings: none.

TypeThreeAssembly.WeightedPerimeter

Source hash: ac2df4f2daf804e358c901d1a6b496eaadc35076745e389cd399733fc5dead10

Relation: equivalent

Source TypeThreeAssembly.lean: 269-272: The perimeter definition directly uses the canonical complete-frontier measure of the actual carrier.

Findings: none.

TypeThreeAssembly.angleTerm

Source hash: 8089feb5fd033f48e23e1ab4579cce22afbbbfde128400123f3b8b2b2a245332

Relation: equivalent

Source TypeThreeAssembly.lean: 289-291: The angle combination is density times exterior angle plus the exposed side angle, matching the article's inverse-trigonometric normalization.

Findings: none.

TypeThreeAssembly.bottomChord

Source hash: 08cc63d3aedc396b5efa9b27fd1bac931beb8b7d14cbce558e5fde942be277c2

Relation: equivalent

Source TypeThreeAssembly.lean: 136-139: The scalar bottom chord is twice the side half-width, with no positivity requirement in the definition.

Findings: none.

TypeThreeAssembly.bottomChord_eq_zero_iff

Source hash: 1b82dffbe772075bcb5095c9dc075644068e9232dc927ae0cd23919b863062dd

Relation: equivalent

Source TypeThreeAssembly.lean: 355-358: Twice the half-width vanishes at exactly the same parameter, giving the stated if-and-only-if degeneracy condition.

Findings: none.

TypeThreeAssembly.bottomChord_nonneg

Source hash: bbe5a0b68c27b1eab70889a99796728aff04d58c47eda0333c16191c78a8f12a

Relation: equivalent

Source TypeThreeAssembly.lean: 352-354: The bottom chord is nonnegative and may vanish; the author does not replace this with unconditional positivity.

Findings: none.

TypeThreeAssembly.bottomChord_pos

Source hash: 5deb1da8ce95e59329d432d4f1484bad3b915566f05bd8a9e6b37a69ad4de177

Relation: equivalent

Source TypeThreeAssembly.lean: 359-362: The non-half-curvature premise is preserved in the strict bottom-chord positivity assertion.

Findings: none.

TypeThreeAssembly.bottomSegmentCarrier

Source hash: 18e70891a1386b33220d27cc96af45d0ffdcc88b970fd467c64035c04da589f5

Relation: equivalent

Source TypeThreeAssembly.lean: 132-135: The closed bottom segment is retained when its half-width vanishes, giving a singleton rather than the empty set.

Findings: none.

TypeThreeAssembly.bottomSegmentCarrier_subset_frontier

Source hash: 82d466e4484f4553cdf0fe29df6958b297f06ff314377fa09237775f676763e1

Relation: equivalent

Source TypeThreeAssembly.lean: 1058-1079: The entire bottom segment lies on the actual frontier, including the zero-length case, because the carrier has no points below the lower interface.

Findings: none.

TypeThreeAssembly.boundaryJoinSet

Source hash: 4d107bd731d106706191e0cc66c1eb6e52b0ad83df797a9379d71e46e9ea68c7

Relation: equivalent

Source TypeThreeAssembly.lean: 1130-1135: All four top and bottom junction coordinates are retained, and repeated bottom points are permitted at zero width.

Findings: none.

TypeThreeAssembly.boundaryJoinSet_finite

Source hash: fb8b2d3307f8905b1616f7f8934468dae8dc2d1df098e8fa72f92f5a3759db5d

Relation: equivalent

Source TypeThreeAssembly.lean: 1136-1141: The explicitly defined junction set is finite, independently of whether its two bottom points coincide.

Findings: none.

TypeThreeAssembly.boundaryTrace

Source hash: 77ded3af26dffe034b843162e20202cd8e868c69330a0cb95819f9e3b2a0bbf5

Relation: equivalent

Source TypeThreeAssembly.lean: 1595-1599: The declared trace contains exactly the two side arcs, upper cap arc and full bottom segment, but not the upper chord interior.

Findings: none.

TypeThreeAssembly.boundaryTrace_subset_frontier_carrier

Source hash: daeda01e39ff76764527644f12850660eec76e1d4031dd8579c24a3b55dda6ba

Relation: equivalent

Source TypeThreeAssembly.lean: 2840-2847: The reverse inclusion establishes that each declared exposed piece belongs to the actual frontier.

Findings: none.

TypeThreeAssembly.carrier

Source hash: 0849c97e398a59f917657f18205fab9ffd6aa8e524ee751befd78f1c4735a4ff

Relation: equivalent

Source TypeThreeAssembly.lean: 262-264: The full carrier is the union of core and upper cap; no lower exterior cap is introduced.

Findings: none.

TypeThreeAssembly.continuous_realizedLeftArcParam

Source hash: 5aa13a8e0a54ece98427e269f77eefa81d89485ad9d1f32dde5c0e6f344ef9b4

Relation: equivalent

Source TypeThreeAssembly.lean: 690-699: The realized left parametrization is continuous globally in its real parameter, not merely on the selected interval.

Findings: none.

TypeThreeAssembly.continuous_realizedRightArcParam

Source hash: 3d9c0740b38c1fa009799b8c35785a0cb767211b94d4ca0bc37637a119a58afd

Relation: equivalent

Source TypeThreeAssembly.lean: 700-709: The realized right parametrization has the same global continuity property.

Findings: none.

TypeThreeAssembly.coreCarrier

Source hash: 5e82c9d38bf8a70ad9c2ae143c293bbf0238ee48148c3dd2d922c95471160878

Relation: equivalent

Source TypeThreeAssembly.lean: 99-103: The actual core is the union of the rectangle and both clipped disks, not a scalar area surrogate.

Findings: none.

TypeThreeAssembly.coreCarrier_y_bounds

Source hash: a0a8c181bd45290d862033f8524f84e76d590a21a370fbbc0214aab783d9df85

Relation: equivalent

Source TypeThreeAssembly.lean: 1447-1455: Every core component imposes both strip height bounds, so the absolute-value conclusion is exact.

Findings: none.

TypeThreeAssembly.core_weightedArea_formula

Source hash: 5a80fac760bbad4e64452fa4d575b54c924b85ffbad6442cb8801547dabcba7

Relation: equivalent

Source TypeThreeAssembly.lean: 1530-1550: The core lies in the density-one branch, so its actual weighted area equals the already computed Euclidean volume.

Findings: none.

TypeThreeAssembly.cos_innerAngle

Source hash: e774d0a2578a71402d098d5ac027f2d6d1dbb90425ca111832168cfb9f013412

Relation: equivalent

Source TypeThreeAssembly.lean: 178-183: The cosine-of-inverse-cosine identity evaluates to the shape parameter on its proved argument domain.

Findings: none.

TypeThreeAssembly.cos_outerAngle

Source hash: 3eb066bbadef1d5d0722d2c322b7cfe8bce8e1eb62204f2872b1f8f1bbb41a24

Relation: equivalent

Source TypeThreeAssembly.lean: 184-188: The corresponding outer-angle cosine evaluates to the density-scaled shape parameter exactly.

Findings: none.

TypeThreeAssembly.cos_sideArcAngle

Source hash: a8f9a8c72dcee4cc4dc59b4da10f2ae6d2741d7865a4b2faca10168fa17cc76a

Relation: equivalent

Source TypeThreeAssembly.lean: 231-237: The cosine changes sign under supplementation, producing the negative shape parameter.

Findings: none.

TypeThreeAssembly.euclidean_frontier

Source hash: 5e03c2ded7f93ebce1b77460290c9d90f4e5be41990287351d94b69c624d60ea

Relation: equivalent

Source TypeThreeAssembly.lean: 2937-2949: The coordinate homeomorphism transports the exact four-piece frontier identity into the Euclidean metric space.

Findings: none.

TypeThreeAssembly.exact_euclidean_bottomSegmentCarrier_hausdorffMeasure

Source hash: 56949577898a4e9f1638a1a8bab2c43a8263ea9e6cf0976fce47fbff56b3fe00

Relation: equivalent

Source TypeThreeAssembly.lean: 1101-1129: The Hausdorff-length identity includes the singleton branch; only the positive-width branch uses the positive-length horizontal-segment primitive.

Findings: none.

TypeThreeAssembly.exact_euclidean_leftArcParam_hausdorffMeasure

Source hash: c6dc496bb93e612df391bdcd13229ada6ddb03a05f7a4a85318498972631a8fc

Relation: equivalent

Source TypeThreeAssembly.lean: 742-778: The exact left image length is radius times side angle, not twice that value; the strict sub-circumference span hypothesis is available.

Findings: none.

TypeThreeAssembly.exact_euclidean_leftArcTrace_hausdorffMeasure

Source hash: 9b484f3710815971495ad59b2a292d470a94ac2f3cad3edb96b8870979733b3

Relation: equivalent

Source TypeThreeAssembly.lean: 817-824: The trace-image equality transfers the actual left parameter-image Hausdorff measure to the coordinate trace's Euclidean image.

Findings: none.

TypeThreeAssembly.exact_euclidean_rightArcParam_hausdorffMeasure

Source hash: f1f573c6c931308699d4611129bc237b3c89fe21d3a82fbb640a2bc4f452f054

Relation: equivalent

Source TypeThreeAssembly.lean: 779-816: The right image has the same single-side length and the correct positive parameter interval.

Findings: none.

TypeThreeAssembly.exact_euclidean_rightArcTrace_hausdorffMeasure

Source hash: 82deac992e3b3149a73b5a9867d35c880f6af8171bd8434cc5a883788eaac184

Relation: equivalent

Source TypeThreeAssembly.lean: 825-832: The right trace has the corresponding exact single-side Hausdorff length.

Findings: none.

TypeThreeAssembly.frontier_carrier

Source hash: 0ce2b73ef2aba31ce763d170404d962fb774d3c494bf5afe10757388adbfbdb30

Relation: equivalent

Source TypeThreeAssembly.lean: 2848-2854: The exact complete-frontier identity follows from both inclusions and underlies the canonical perimeter computation.

Findings: none.

TypeThreeAssembly.frontier_carrier_subset_boundaryTrace

Source hash: e4a9f2900fa53b9204f0a28ca553872a35240978918b3cf32712160073f759cb

Relation: equivalent

Source TypeThreeAssembly.lean: 2443-2458: The frontier inclusion is derived from the coordinate union, excluding internal seams rather than assuming the desired boundary decomposition.

Findings: none.

TypeThreeAssembly.innerAngle

Source hash: bb9acabc2bc9f5427d7041305c401b4e2793720759dd063e3153448db0793ddb

Relation: equivalent

Source TypeThreeAssembly.lean: 53-55: The inner angle is principal inverse cosine of the shape parameter, not the exposed side-arc turning angle.

Findings: none.

TypeThreeAssembly.innerAngle_lt_pi

Source hash: 31385b2dc48409bb1698d05b2b8cedcea6151f9ce4bedc38534228b77f838b5c

Relation: equivalent

Source TypeThreeAssembly.lean: 168-170: The strict lower shape bound keeps the inner angle strictly below a half-turn.

Findings: none.

TypeThreeAssembly.innerAngle_pos

Source hash: e78599fffc8793999ca239c142df39a815df5b9220a19bd2bd882d169b85658b5

Relation: equivalent

Source TypeThreeAssembly.lean: 165-167: The strict upper shape bound implies a strictly positive principal inner angle.

Findings: none.

TypeThreeAssembly.innerAngle_semicircular

Source hash: 136b95db3cfbe6ffba49a9d377637e6fd44e92cfd37653b368ff453e7d5af642

Relation: equivalent

Source TypeThreeAssembly.lean: 370-373: At half curvature the shape parameter vanishes, giving the stated principal inner right angle.

Findings: none.

TypeThreeAssembly.innerSin_pos

Source hash: 88694b992a38f1dff3c67f32d082f4eb742c98f01874e62941887bfa3c13f04d

Relation: equivalent

Source TypeThreeAssembly.lean : 216-218: The same open principal range gives positive inner sine, as required for the auxiliary cap chord.

Findings: none.

TypeThreeAssembly.integrableOn_carrier

Source hash: f50bce5009ef96b4081db29960a7e72186fb76edbfedf9fa6fd538bc59ab8c22

Relation: equivalent

Source TypeThreeAssembly.lean : 3000-3011: Finite core volume and finite cap volume give density integrability over the actual union, without assuming its area formula as input.

Findings: none.

TypeThreeAssembly.integrable_frontierMeasure

Source hash: 242285e9e93302048576853114a4bc7f6ca16cdc917f22c33a2fe025def880a8

Relation: equivalent

Source TypeThreeAssembly.lean : 3012-3042: Exact finite Hausdorff lengths of all frontier pieces give finite canonical mass and integrability of the bounded density, including zero bottom length.

Findings: none.

TypeThreeAssembly.isClosed_bottomSegmentCarrier

Source hash: 4502cfa8cbb9cfeeb9921da9cbccfe0b4a2a7c969d9473aede4beb6ee9ada554

Relation: equivalent

Source TypeThreeAssembly.lean : 1589-1594: The bottom segment is closed even when it is a singleton; no positive-width premise is added.

Findings: none.

TypeThreeAssembly.isClosed_carrier

Source hash: 86d49c14fed38ff9d123ec7002dc853d7ab13d112817308f4d1ef5ac7d463980

Relation: equivalent

Source TypeThreeAssembly.lean : 1307-1309: The full carrier is a union of the closed core and closed upper cap, with no extra regularity premise.

Findings: none.

TypeThreeAssembly.isClosed_coreCarrier

Source hash: c084bdc57b0172b2f03efa14200c62e0c7ef9b7e0265bac3ea60cec5c75fc390

Relation: equivalent

Source TypeThreeAssembly.lean: 1303-1306: The actual core is a finite union of the three closed primitive pieces.

Findings: none.

TypeThreeAssembly.isClosed_leftSegmentCarrier

Source hash: 9a06aa24438ad23f7c7f01c4c87ea6fc01e48b3ecec4d358799e726bea34abb1

Relation: equivalent

Source TypeThreeAssembly.lean: 1289-1295: The left disk and each clipping half-plane are closed, yielding the stated carrier closedness.

Findings: none.

TypeThreeAssembly.isClosed_rectangleCarrier

Source hash: a040fbd7853bc5d56529ed0d45def53a17db2084e9e6f891db5067b23f32a560

Relation: equivalent

Source TypeThreeAssembly.lean: 1282-1288: All four rectangle coordinate constraints are closed, including the zero-width rectangle.

Findings: none.

TypeThreeAssembly.isClosed_rightSegmentCarrier

Source hash: 97ec89af687cf23451a62fdd3fb4816f90150533b2e6c5aca3fc7cb1fe8e300f

Relation: equivalent

Source TypeThreeAssembly.lean: 1296-1302: The right disk has the reflected closed cutoff and the same closed strip bounds.

Findings: none.

TypeThreeAssembly.isometry_sideArcComplexIsometryMap

Source hash: 5f9f98564eebaed09c3cd958295e743ceabd92f2b665793225e0fb64c4812ba4

Relation: equivalent

Source TypeThreeAssembly.lean: 653-664: The arbitrary-center map preserves Euclidean distances; the author does not restrict it to an assembly center.

Findings: none.

TypeThreeAssembly.leftArcParam

Source hash: 4f29c060ae66edf060cbefb1375e17d2584dc8c159343c7bce4a1ff009d7632c

Relation: equivalent

Source TypeThreeAssembly.lean: 120-126: The sine horizontal displacement and negative cosine vertical displacement agree; the selected interval correctly ends at bottom tangency.

Findings: none.

TypeThreeAssembly.leftArcParam_image_Icc

Source hash: a53d3bd58a5cf0ca117ca4045079faea921dc67e709030e1bc93b3bc04963479

Relation: equivalent

Source TypeThreeAssembly.lean : 446-546: The exact left parameter image has the asymmetric interval ending at zero, not the full symmetric auxiliary cap interval.

Findings: none.

TypeThreeAssembly.leftArcTrace

Source hash: cfb74d8a54830ee4e695b37318a8d8c8a086063cc569f3e8d4fa167da3d5d0b9

Relation: equivalent

Source TypeThreeAssembly.lean : 104-110: The left circle equality and three cutoff inequalities give the exact exposed trace, including endpoints.

Findings: none.

TypeThreeAssembly.leftCenter

Source hash: ae634544d50a625f2da0ab83dddef536c2004aefe8a002e19ec3ff792534bdf2

Relation: equivalent

Source TypeThreeAssembly.lean : 70-72: The left center has negative half-width and vertical coordinate radius minus one, giving bottom tangency.

Findings: none.

TypeThreeAssembly.leftSegmentCarrier

Source hash: 7a6b1fb5a536fc1614b926138e279c79369045c2b292e00c145e1a0e699dbced

Relation: equivalent

Source TypeThreeAssembly.lean : 85-91: The left disk is clipped at its center line and both strip interfaces, with all inequalities non-strict.

Findings: none.

TypeThreeAssembly.leftUpperJunction

Source hash: a660a79d351317f739116ce079a065c0c8bcd856fae9b73442ad8d1339dd7ec

Relation: equivalent

Source TypeThreeAssembly.lean : 946-957: The side-width identity makes the left core top junction exactly equal to the outer cap's left endpoint.

Findings: none.

TypeThreeAssembly.measurableSet_carrier

Source hash: 25696aa660c5c71e8ce3e6a179a069d25633bc99ff664ad8e93ea6e319cb501f

Relation: equivalent

Source TypeThreeAssembly.lean : 1310-1316: Closedness implies Borel measurability of the actual coordinate carrier.

Findings: none.

TypeThreeAssembly.measurableSet_euclidean_bottomSegmentCarrier

Source hash: ac873340c9031353c9d26e2404b771c8a0ff1b638ea75b1a24ed627ed5bc802d

Relation: equivalent

Source TypeThreeAssembly.lean: 2855-2860: The Euclidean bottom-segment image is measurable also in the singleton case, as required by four-piece integral additivity.

Findings: none.

TypeThreeAssembly.measurableSet_euclidean_leftArcTrace

Source hash: 791dc1e2fe6e2d0cb10c51a3bd4f4018c5ee53a25eeef24b22bc0d4e6f0f1192

Relation: equivalent

Source TypeThreeAssembly.lean: 732-736: The Euclidean left trace is a compact continuous parameter image and hence measurable.

Findings: none.

TypeThreeAssembly.measurableSet_euclidean_rightArcTrace

Source hash: 64197e0a340a1643ad8bc1525fe961f5b8792a0e7cb9321824fa2436e9d02bb4

Relation: equivalent

Source TypeThreeAssembly.lean: 737-741: The same compact-image argument yields measurability of the Euclidean right trace.

Findings: none.

TypeThreeAssembly.modeledWeightedArea

Source hash: 35499fdca0a3bbbed11909d6dec0bbb1c83123d6ea5cb0599717ee289f567e87e

Relation: equivalent

Source TypeThreeAssembly.lean: 278-284: The modeled area contains the rectangle, auxiliary side cap and density-weighted outer cap, with equality to the actual integral proved later.

Findings: none.

TypeThreeAssembly.modeledWeightedArea_formula

Source hash: b0e70e8aeafad1e18dbaff1d3d6364f544c6042f48ce97bebd3fa5c6cc8e74e

Relation: equivalent

Source TypeThreeAssembly.lean: 885-895: The modeled area is algebraically identified with the scalar formula before the actual-integral bridge; the author preserves that distinction.

Findings: none.

TypeThreeAssembly.modeledWeightedPerimeter

Source hash: 76198be6eb5b8790a4af34449bf29c4b963314e1e4bd268b4d3c9cdb2e69ea0d

Relation: equivalent

Source `TypeThreeAssembly.lean`: 285-288: The modeled perimeter includes both strip arcs through the auxiliary cap length, the weighted outer arc and the full bottom chord.

Findings: none.

TypeThreeAssembly.modeledWeightedPerimeter_formula

Source hash: bb4c05a2ca2fd9d5b63be1b6ad16da716cc43497abcad228054765e6eaa2e39e

Relation: equivalent

Source `TypeThreeAssembly.lean`: 896-903: The modeled perimeter simplifies to the scalar expression with the bottom segment included, not discarded as an attaching chord.

Findings: none.

TypeThreeAssembly.one_lt_radius

Source hash: 4e892eef63bc8252d5a243e2b5495e748b55fa956d516b3510c9e5289ec264d9

Relation: equivalent

Source `TypeThreeAssembly.lean`: 148-152: The strict upper curvature bound yields a strict lower radius bound, excluding the three-arc endpoint.

Findings: none.

TypeThreeAssembly.outerAngle

Source hash: 0e2ff2070394fdda5b4ec09824f47ed543dabdfef101c076ad378e0c92be2ff47

Relation: equivalent

Source `TypeThreeAssembly.lean`: 59-63: The principal inverse-cosine definition permits the full open angle range, retaining the major exterior cap branch.

Findings: none.

TypeThreeAssembly.outerAngle_lt_pi

Source hash: ce135b6a3dc191fb33efd487aab9da31365b8ec038647a624b62a436ba911d34

Relation: equivalent

Source `TypeThreeAssembly.lean`: 174-177: The strict outer-argument lower bound excludes a full-circle cap traversal with coincident endpoints.

Findings: none.

TypeThreeAssembly.outerAngle_pos

Source hash: 760f9b90fd549c278aa760178ec027d7c643f8b5bd6d4bf942c86f6a65bb0e28

Relation: equivalent

Source `TypeThreeAssembly.lean`: 171-173: The strict outer-argument upper bound excludes a zero exterior half-angle.

Findings: none.

TypeThreeAssembly.outerAngle_semicircular

Source hash: 9ff26a1092e7089058ecf98f24fecce78972031b475870c0e4d30009e831f8ee

Relation: equivalent

Source TypeThreeAssembly.lean: 374-377: The outer argument also vanishes at half curvature, giving the same exterior half-angle for every admissible density.

Findings: none.

TypeThreeAssembly.outerArgument

Source hash: d6cfc025aa1368c77ccefa6b050dbf8bfa113c35569b4434625ec2d735b91f41

Relation: equivalent

Source TypeThreeAssembly.lean: 56-58: The outer cosine argument is the shape parameter divided by density, with the same denominator placement.

Findings: none.

TypeThreeAssembly.outerArgument_mem

Source hash: fc2a8b1032448c01ece41eff0bcd5d748f70ee556d5721867bb36b59d56ce63a

Relation: equivalent

Source TypeThreeAssembly.lean: 156-164: The density jump gives a positive denominator larger than one, so both strict outer-argument bounds hold.

Findings: none.

TypeThreeAssembly.outerCap

Source hash: 5482055613f2ec0454ddd9c71d722bea04494543bc659e447732870c662eb86f

Relation: equivalent

Source TypeThreeAssembly.lean: 251-261: The upper cap has the radius-derived positive chord and the actual outer half-angle, with the correct upper orientation.

Findings: none.

TypeThreeAssembly.outerCap_arcLength

Source hash: e4c6a09e492a4c9395262e1c277f2cd0b185e72ae34042a7f14138b12c6b4c19

Relation: equivalent

Source TypeThreeAssembly.lean: 875-884: The outer selected arc length is radius times the full central angle, with the factor two retained.

Findings: none.

TypeThreeAssembly.outerCap_center

Source hash: 1057a7a30ad6e7558c1c0e586fc0147d20f1fc01e6c3e508acda901e55191c4c

Relation: equivalent

Source TypeThreeAssembly.lean: 929-933: Expansion of the outer cap center and its radius identity yields exactly the declared upper center.

Findings: none.

TypeThreeAssembly.outerCap_euclideanArea

Source hash: 3c7709918bd2e08d7c7ea431b339d946f2eb6c3c9d1138f4304b6afe64a153c2

Relation: equivalent

Source TypeThreeAssembly.lean: 851-864: The outer scalar cap area uses the outer half-angle on its full admissible range, including major segments.

Findings: none.

TypeThreeAssembly.outerCap_leftEndpoint

Source hash: 369e532ae6bbe742ee13471a1a8cc638a185e8c74c4f29cdde432417763b0fdb

Relation: equivalent

Source TypeThreeAssembly.lean: 934-938: The left upper endpoint is at the negative outer half-chord on the upper interface.

Findings: none.

TypeThreeAssembly.outerCap_radius

Source hash: f107696210fe89f5a746d54584303c70da482f05587bc229f2d37aaa9494b57f

Relation: equivalent

Source TypeThreeAssembly.lean: 419-422: The outer chord definition and positive sine cancel to give the assembly radius exactly.

Findings: none.

TypeThreeAssembly.outerCap_rightEndpoint

Source hash: 0bf0b00b58f0396d9b9b8bc9e08617bcd76cf3f0b76937e23b52dec1f9b5fe12

Relation: equivalent

Source TypeThreeAssembly.lean: 939-945: The right upper endpoint is at the positive outer half-chord on the same interface.

Findings: none.

TypeThreeAssembly.outerSin_pos

Source hash: feedd827e6203dedcb21f4f474b9fea36ae3ea708b33b644b944093c5f958959

Relation: equivalent

Source TypeThreeAssembly.lean: 213-215: The outer angle lies strictly between zero and a half-turn, giving strictly positive sine.

Findings: none.

TypeThreeAssembly.pairwiseOverlap_h1_null

Source hash: 545b8d3e4922237126e7364ee9bb07b444cf8cada72b02e8afca89501d2f79b3

Relation: equivalent

Source TypeThreeAssembly.lean: 1258-1273: The Euclidean overlap measure is zero by containment in the finite junction image, providing the needed almost-disjointness.

Findings: none.

TypeThreeAssembly.pairwiseOverlap_subset_boundaryJoinSet

Source hash: f8736f4dd5a67fc4149e15b1ed3e3f21030a49e12643cd517b41158566d4664f

Relation: equivalent

Source TypeThreeAssembly.lean: 1142-1257: The overlap inclusion considers all six pairs of frontier pieces and confines them to the stated finite junctions, including the degeneracy.

Findings: none.

TypeThreeAssembly.radius

Source hash: 44bdb144ee33a0fe634670e914d151b3f9a50f2b2f7db8da7c3fad23bdfb70e

Relation: equivalent

Source TypeThreeAssembly.lean: 47-49: The radius is the reciprocal of the positive curvature, and the strict upper curvature bound gives radius greater than one.

Findings: none.

TypeThreeAssembly.radius_mul_h

Source hash: e90d4095fef5e1e089527a179a53736b5dbf462b6149abecb5375e96b7380078

Relation: equivalent

Source TypeThreeAssembly.lean: 197-200: The positive curvature is nonzero, so multiplication by its reciprocal gives the stated unit product.

Findings: none.

TypeThreeAssembly.radius_pos

Source hash: a610fc617d9a92f1154e4b9627bb16a0d909464945f9e22ccb900c975acb2c7d

Relation: equivalent

Source TypeThreeAssembly.lean: 145-147: Positive curvature gives positive reciprocal radius, exactly as stated.

Findings: none.

TypeThreeAssembly.realizedLeftArcParam

Source hash: 51fbf9517aa10415de53dc106db635c5f70b9c39860ba75778e696cdeb2d6c3

Relation: equivalent

Source TypeThreeAssembly.lean: 665-668: The realized left arc is obtained by the coordinate homeomorphism with no alteration of the curve's point coordinates.

Findings: none.

TypeThreeAssembly.realizedLeftArcParam_eq

Source hash: 538a7925bd9a6977d5e74933bd5f951885abb82dc534e3d102db01bf6aa01a1e

Relation: equivalent

Source TypeThreeAssembly.lean: 672-680: The realized left curve agrees pointwise for every real parameter with the isometric image of the standard circle about the left center.

Findings: none.

TypeThreeAssembly.realizedLeftArcParam_image_Icc

Source hash: 06e6662a39cd836487840418c7224db9c1b8779c37aa295aab4e1109503a76c7

Relation: equivalent

Source TypeThreeAssembly.lean: 710-720: Transporting the exact coordinate image gives the Euclidean left trace on the correct negative-to-zero interval.

Findings: none.

TypeThreeAssembly.realizedRightArcParam

Source hash: 8a963395af5bf096a9c5748471f90364728d8fe511c4e2cc0591bd2068d9e016

Relation: equivalent

Source TypeThreeAssembly.lean: 669-671: The realized right arc uses the same Euclidean transport and the right coordinate parametrization.

Findings: none.

TypeThreeAssembly.realizedRightArcParam_eq

Source hash: 3339fa9296e8c62289cd021c3334658bb86b73401d06cffbaa790d7423ffd93e

Relation: equivalent

Source TypeThreeAssembly.lean: 681-689: The realized right curve has the corresponding pointwise isometric representation about the right center.

Findings: none.

TypeThreeAssembly.realizedRightArcParam_image_Icc

Source hash: 9469dc298ecad88323016b9907111e322a864f0831a991b637ca11824881af16

Relation: equivalent

Source TypeThreeAssembly.lean: 721-731: The corresponding positive interval transports exactly onto the Euclidean right trace.

Findings: none.

TypeThreeAssembly.rectangleArea

Source hash: 3314187b7b48ab958a7d8415bcf10afdb2dd85317d354dfd049f5344c2569e90

Relation: equivalent

Source TypeThreeAssembly.lean: 273-277: The rectangle area is its height times twice the half-width and remains valid for zero width.

Findings: none.

TypeThreeAssembly.rectangleCarrier

Source hash: d451775e4018b5bae444c6a8ed6911c5adcc23db4a0efc68ab71c655f2933477

Relation: equivalent

Source TypeThreeAssembly.lean: 80-84: The closed rectangle bounds agree and permit zero width; the constructor does not require a nondegenerate rectangle.

Findings: none.

TypeThreeAssembly.rightArcParam

Source hash: 74515bd4fc474dfa2c52a14662b8cc34aa0f40ef9d8185819e05d1de94ccf588

Relation: equivalent

Source TypeThreeAssembly.lean: 127-131: The right parametrization has the same orientation formula around the reflected center and starts at bottom tangency on its selected interval.

Findings: none.

TypeThreeAssembly.rightArcParam_image_Icc

Source hash: d47f88fb5eadc69e4f4bc7ce3bc201dc34d60ad570f34d3d29c4d0184ef0f8f1

Relation: equivalent

Source TypeThreeAssembly.lean: 547-648: The right parameter image uses the interval starting at zero and ending at the positive side-arc length.

Findings: none.

TypeThreeAssembly.rightArcTrace

Source hash: 828cebab0c56e67a2a3cfaae9e912a8a76f497287dcea6b589e1a22a40edd616

Relation: equivalent

Source TypeThreeAssembly.lean: 111-119: The right circle equality and reflected cutoff give the corresponding closed exposed trace.

Findings: none.

TypeThreeAssembly.rightCenter

Source hash: c1d50740a1dfb76a9c424f19f6e78589f8a8ab930ab1727da5282aec42c3f25

Relation: equivalent

Source TypeThreeAssembly.lean: 73-75: The right center has positive half-width and the same vertical coordinate, preserving reflection symmetry.

Findings: none.

TypeThreeAssembly.rightSegmentCarrier

Source hash: 889596cedc4d7d957005581540478e3a458f9f3d1f37c9af6b1a713eee05b078

Relation: equivalent

Source TypeThreeAssembly.lean: 92-98: The right disk uses the reflected horizontal cutoff and the same closed strip clipping.

Findings: none.

TypeThreeAssembly.rightUpperJunction

Source hash: ae6c6a14cc5b76c0f3aa60155a1fb16c629e4d67ae173c7593aa69c135bbff39

Relation: equivalent

Source TypeThreeAssembly.lean: 958-968: The reflected identity gives exact equality of the right attaching junctions.

Findings: none.

TypeThreeAssembly.scalarWeightedArea

Source hash: 195fdb382725116907cc721bab689c5c6f6f34b356a442f900ba649d89b62b9c

Relation: equivalent

Source TypeThreeAssembly.lean: 295-299: The scalar area retains the shape-plus-two coefficient and squared-curvature denominator.

Findings: none.

TypeThreeAssembly.scalarWeightedPerimeter

Source hash: fcbac6602a2b133a8e714c37e1ef51bfb698ab0ae754dcd7eca0cb6e9c49204d

Relation: equivalent

Source TypeThreeAssembly.lean: 300-302: The scalar perimeter retains the factor two, the angle-plus-gap numerator and the first-power curvature denominator.

Findings: none.

TypeThreeAssembly.shapeParameter

Source hash: 29190b0ec0676db3ee8da889fc7b1e078bce75188f6b4271945e3fee045f2920

Relation: equivalent

Source TypeThreeAssembly.lean: 50-52: The affine shape definition agrees and belongs to the open inverse-cosine domain under the constructor bounds.

Findings: none.

TypeThreeAssembly.shapeParameter_mem

Source hash: 018fe2d07fb99b9420ec19a75327dc616f0119968b1c22fb83265de47b1261df

Relation: equivalent

Source TypeThreeAssembly.lean: 153-155: Both strict shape-parameter bounds follow from the open curvature interval and are retained.

Findings: none.

TypeThreeAssembly.sideArcAngle

Source hash: c68849312591ff361810874588f817eab8327b61ad9b815244ad1ab61a55597a

Relation: equivalent

Source TypeThreeAssembly.lean: 64-65: The exposed side-arc angle is the supplement of the inner angle; the auxiliary symmetric cap has twice that central angle.

Findings: none.

TypeThreeAssembly.sideArcAngle_lt_pi

Source hash: 27c4b01d9c8710ab675792367c9a16f3b7952720c452f471c97b46d487e08c08

Relation: equivalent

Source TypeThreeAssembly.lean: 223-226: The positive inner angle makes its supplement strictly smaller than a half-turn.

Findings: none.

TypeThreeAssembly.sideArcAngle_pos

Source hash: fb3e4cdd6ff568be2285a238c6293705a5d18ec17ca0000cf9fc5fd3be6f53e2

Relation: equivalent

Source TypeThreeAssembly.lean: 219-222: The supplement of the strictly sub-half-turn inner angle is strictly positive.

Findings: none.

TypeThreeAssembly.sideArcComplexIsometryMap

Source hash: 2d2b0ae6442b1aeb7779e0e2c761b53071a3b1948418645b546ba443dcc3b322

Relation: equivalent

Source TypeThreeAssembly.lean: 649-652: The map is defined for an arbitrary coordinate center and applies the exact coordinate interchange, sign and translation stated.

Findings: none.

TypeThreeAssembly.sideCap

Source hash: 62a5fb2c6003ebc71ae1cea156d819929dd2061bdfd35d03ac33c3bf896e8fd0

Relation: equivalent

Source TypeThreeAssembly.lean: 238-250: The auxiliary cap is based at the upper interface with lower orientation and the prescribed full side angle; it is not an extra competitor component.

Findings: none.

TypeThreeAssembly.sideCap_arcLength

Source hash: 0735084a84605f457abd29f126bfc3b41f13fe2e361355b34befa71a89121ae9

Relation: equivalent

Source TypeThreeAssembly.lean: 865-874: The auxiliary side cap length is twice one side-arc length and therefore represents both exposed strip arcs together.

Findings: none.

TypeThreeAssembly.sideCap_center

Source hash: 5c32cc16475c4c5293c8fcbce19421c65fe608e9916172122a7cb5ff16d6b389

Relation: equivalent

Source TypeThreeAssembly.lean: 430-438: The lower-oriented auxiliary cap center simplifies to the radius-minus-one height using the radius-curvature product.

Findings: none.

TypeThreeAssembly.sideCap_euclideanArea

Source hash: b6b94a03ab653ee46e011e7fa757125fdbdfe26f1473cd6911d9100fe92583e4

Relation: equivalent

Source TypeThreeAssembly.lean: 833-850: The auxiliary cap's scalar area has the standard radius-squared segment formula at the supplementary angle.

Findings: none.

TypeThreeAssembly.sideCap_radius

Source hash: 860b391d38da56536b3934963761766a28238619240605477033823fc900d1af

Relation: equivalent

Source TypeThreeAssembly.lean: 423-429: The auxiliary side-cap radius agrees because supplementary angles have equal positive sines.

Findings: none.

TypeThreeAssembly.sideHalfWidth

Source hash: c1a6374b950f6fdf60b58062fd5753e6cafd953a8fbac46cb6164191b66af1b3

Relation: equivalent

Source TypeThreeAssembly.lean: 66-69: The width is radius times outer-minus-inner sine, in the correct order; the bottom half-length interpretation matches the later carrier theorem.

Findings: none.

TypeThreeAssembly.sideHalfWidth_eq_zero_iff

Source hash: aa31042682af8c9fc65e3a19a4a9ce7dc71ed7e2ad3e807732dcfcc2da35644f

Relation: equivalent

Source TypeThreeAssembly.lean: 344-351: The formal equivalence identifies the exact zero-width locus, with equality at one half and strict positivity everywhere else in the assembly domain.

Findings: none.

TypeThreeAssembly.sideHalfWidth_nonneg

Source hash: be70cb71b25ed2f43754461915097af9183cc7aac3504655dda98d8a67b605ec

Relation: equivalent

Source TypeThreeAssembly.lean: 338-340: Nonnegative sine gap times positive radius gives a nonnegative bottom half-width, including the disk branch.

Findings: none.

TypeThreeAssembly.sideHalfWidth_pos

Source hash: a769284707d6bf25198d747cd5846bae22f2f394cc9d09e601cfcab854579d7a

Relation: equivalent

Source TypeThreeAssembly.lean: 341-343: The author retains the non-half-curvature premise needed for strictly positive width.

Findings: none.

TypeThreeAssembly.sin_inner_le_sin_outer

Source hash: def468198882d033570072298a83ae1d41fc003f0ecb9d0b4ee3a6b94e6a1d10

Relation: equivalent

Source TypeThreeAssembly.lean: 303-311: The non-strict sine comparison follows from comparing the two nonnegative square roots under the density jump.

Findings: none.

TypeThreeAssembly.sin_inner_lt_sin_outer

Source hash: b3a6348544f17baf0cf9a8d51663f5b8b6187359294117163fc5d2afba78dd58

Relation: equivalent

Source TypeThreeAssembly.lean: 312-337: The exclusion of the half-curvature parameter is explicitly retained and is exactly what makes the radical comparison strict.

Findings: none.

TypeThreeAssembly.sin_sideArcAngle

Source hash: 41165bd4a88985079154632e1304282b8868bb98cc6b1d97bb3ea19fae0dd

Relation: equivalent

Source TypeThreeAssembly.lean: 227-230: The sine of the supplementary side angle equals the inner sine, with no sign change.

Findings: none.

TypeThreeAssembly.sineGap

Source hash: e9d7a7074e27fe28a23aa0db9c4cf9b0ef4e5f2181f1cfab6098f7cc2e240ecb

Relation: equivalent

Source TypeThreeAssembly.lean: 292-294: The sine gap uses outer-minus-inner order and is later identified with the article's radical difference.

Findings: none.

TypeThreeAssembly.upperCenter

Source hash: 6dabb1e2d2dfef1af758b047dfb6728a97191f4c64742b75a35f102c81c1c3c1

Relation: equivalent

Source TypeThreeAssembly.lean: 76-79: The upper center has zero horizontal coordinate and the stated cosine offset from the upper interface.

Findings: none.

TypeThreeAssembly.volume_coreCarrier

Source hash: e422839d8a82e578aa8619bf8a6d240c5dc840fd4d4e1880ecc09a0307558380

Relation: equivalent

Source TypeThreeAssembly.lean: 1481-1529: The actual core volume is the rectangle area plus the auxiliary side-cap area after removing planar-null decomposition overlaps.

Findings: none.

TypeThreeAssembly.weightedArea_eq_modeledWeightedArea

Source hash: 5f7913f0171c133683a735330062b186a060d0e518987c5488549de7bfd06347

Relation: equivalent

Source TypeThreeAssembly.lean: 1562-1583: The actual carrier area integral equals the modeled sum because the upper cap and core overlap only on a planar-null interface.

Findings: none.

TypeThreeAssembly.weightedArea_formula

Source hash: 53616c2c25615a95257aa4773c87d4c69464dc905650e6ed5b53696dc5f08570

Relation: equivalent

Source TypeThreeAssembly.lean: 1584-1588: Composing the actual-to-modeled and modeled-to-scalar identities gives the advertised actual scalar area formula.

Findings: none.

TypeThreeAssembly.weightedPerimeter_eq_modeledWeightedPerimeter

Source hash: 3dd334c8f3cff9146f23a5d90e6da2520151c567d49c8604446b90aa7a9f4d2b

Relation: equivalent

Source `TypeThreeAssembly.lean`: 2950-2993: The proof transports the canonical integral, applies exact frontier decomposition and null-overlap additivity, and evaluates the density-one bottom contribution explicitly.

Findings: none.

TypeThreeAssembly.weightedPerimeter_formula

Source hash: c6f51109b06f9f61cbbcb63e9d6e1d950e5a79c8decdf76b0e2fc3f366f8eca

Relation: equivalent

Source `TypeThreeAssembly.lean`: 2994-2999: The actual-to-modeled perimeter bridge and algebraic scalar identity compose to the exact canonical perimeter formula.

Findings: none.

WeightedArea

Source hash: 959b5ddd7363861d4d0d79c211781b3b19e09df50087472690ae0581725b8c6d

Relation: equivalent

Source `CMVGeometry.lean`: 55-60: The definition is the ambient Lebesgue set integral, with integrability proved separately for the geometric carriers.

Findings: none.

WeightedPerimeter

Source hash: 8d1f1a60a6d4c5ce03ccda31c1bc34ec4f837b6ce55275e4b0c06fe04e566dba

Relation: equivalent

Source `CMVGeometry.lean`: 67-73: The functional accepts a measure, and the explanation correctly requires the canonical frontier measure in the final geometric theorem.

Findings: none.

area

Source hash: 8c44e8dff416b2208afe91aad54bf1678ad6f2299c75592fe1fed4f6cbf55dc2

Relation: equivalent

Source `MorgansArcFunction.lean`: 92-93: The normalized area has the exact sine-cosine numerator and squared-sine denominator, including the major-segment geometric interpretation.

Findings: none.

area_pos

Source hash: 0af41b50988d0bace92fc2e638571302802d8538245a2c254bedb9f164463ede

Relation: equivalent

Source `MorgansArcFunction.lean`: 148-157: The strict angle hypotheses make both the numerator and denominator positive, yielding the stated strict normalized area positivity.

Findings: none.

capBoundaryMeasure

Source hash: 5dbd12f9bc57860ad2ce92a638514e0e8d15006685aca7388e2fa2e1689504cd

Relation: equivalent

Source `CMVGeometry.lean`: 1249-1251: The auxiliary measure pushes forward the closed arclength interval and represents the arc only, not the cap's chord.

Findings: none.

capBoundaryMeasure_integrable

Source hash: 0065afd47de261467dd963dc3f68bd9fc5e9eab2c424cfffb1d00f2e7c2919a7

Relation: equivalent

Source `CMVGeometry.lean`: 2260-2267: The cap parametrization satisfies the pushforward integrability hypotheses; the claim concerns the auxiliary arc measure.

Findings: none.

capComplexIsometryMap

Source hash: 3ed78deaf1597a91c6218ca8fe10519ba693355e7329e0301f17fcce226f5a43

Relation: equivalent

Source `FrontierPerimeter.lean`: 390-397: Both orientation branches have the correct coordinate interchange, translation and vertical sign, matching the cap parametrization.

Findings: none.

capEnd

Source hash: a140ca726638df619aef507f8019eb9f7ca9503d3231f8bfa8f0a64feb377161

Relation: equivalent

Source `CMVGeometry.lean`: 1246-1248: The final parameter is the positive radius-half-angle product, so the interval span is the selected arc length.

Findings: none.

capParam

Source hash: d80d6cd5283176fe8b29770f039d1135a8892539144575a4bc1d9962ad427168

Relation: equivalent

Source `CMVGeometry.lean`: 1241-1243: Dividing the arclength parameter by the cap radius yields the angle parameter; the stated selected interval is consistent with this scaling.

Findings: none.

capParam_image_Icc_eq_arcTrace

Source hash: f412e9294ab946fde4ef494447fa75b0b57b0600c2bd61007639df1bb35c0e9e

Relation: equivalent

Source `FrontierPerimeter.lean`: 947-1070: The coordinate image theorem covers the full cap-angle range, so inverse-trigonometric reconstruction does not silently discard major-arc points.

Findings: none.

capPartitionPoint

Source hash: 49ac6ecbe67b9040843eeee672d7b0f08055016781814c3ce0fa4e7d31ff805a

Relation: equivalent

Source `FrontierPerimeter.lean`: 154-156: The partition-point definition is total for all natural counts and indices; positivity is imposed only in subsequent lemmas.

Findings: none.

capPartitionPoint_dist

Source hash: 6b90645e53e16e3855618b966b04af3ee0f15439043926ba1133c583e65b5a00

Relation: equivalent

Source `FrontierPerimeter.lean`: 185-207: The exact consecutive chord distance holds for every natural index, not only those inside the chosen finite partition; the sine sign is justified by the angle bounds.

Findings: none.

capPartitionPoint_end

Source hash: b5d0e986af4c921461bd13ce89490a41fdc02991eaadda7d26f70db465413519

Relation: equivalent

Source `FrontierPerimeter.lean`: 161-166: The strictly positive natural subdivision count is retained because cancellation is needed to identify the final partition point.

Findings: none.

capPartitionPoint_mem

Source hash: 8f26d09621c071ac8a6a1935a21253dfb29cb269e0ec21dc71e59a40f0529e9f

Relation: equivalent

Source `FrontierPerimeter.lean`: 175-184: The bounded-index hypothesis is retained, and monotonicity places each selected point in the closed symmetric interval.

Findings: none.

capPartitionPoint_zero

Source hash: 1714782661300a21ee2d4243c7e4fbe79e1e97ce545ea187699c13d709db2db9

Relation: equivalent

Source `FrontierPerimeter.lean`: 157-160: The initial partition point is independent of the subdivision count, including zero.

Findings: none.

capStart

Source hash: 594eb6e2bbf3cbc7a2397b6af7a93506027168fa75ecf7f70b7012b2fc6e208a

Relation: equivalent

Source `CMVGeometry.lean`: 1244-1245: The initial parameter is the negative radius-half-angle product, retaining the left endpoint.

Findings: none.

cap_arcLength_eq_two_radius_theta

Source hash: edad6e24948a1499ce1a66a5efdc84d9c8f7899f5c52d6551716312e95163fb3

Relation: equivalent

Source `FrontierPerimeter.lean`: 451-456: The scalar chord-normalized arc length equals twice radius times half-angle by nonzero-sine cancellation.

Findings: none.

cap_integrableOn

Source hash: bbc4a1a9af0d45bee6aa1c1152d956e0e945a869f3550ca289ef7022556292ef

Relation: equivalent

Source `CMVGeometry.lean`: 2303-2308: The finite cap volume supplies the integrability premise for the bounded, measurable strip density.

Findings: none.

cap_lower_external_density

Source hash: baaf83c56acf8edc18268269454b4946af9c7098831da004f1da856c3f80d9b2

Relation: equivalent

Source `CMVGeometry.lean`: 1341-1352: The author retains lower orientation and the lower interface base; interior arc points are below that interface.

Findings: none.

cap_lower_weightedArea

Source hash: af4bcab3c0cdacd24a9ba22f365ac164c96fd2be59faf0a944ca162de8e4ada2

Relation: equivalent

Source `CMVGeometry.lean`: 2360-2389: Lower orientation and base at the lower interface are retained, with the exterior density applying almost everywhere in area.

Findings: none.

cap_lower_weightedIntegral

Source hash: 5575ee03ce1e095004c86962659c2bcf1dde6c0b1ad90b439220173f4e11e7fc

Relation: equivalent

Source `CMVGeometry.lean`: 1373-1394: The lower exterior arc has the same weighted-length rule, with both endpoint exceptions correctly treated as null.

Findings: none.

cap_partition_polygon_lower

Source hash: 4dc7fcd936b6c56a4643f00c8d67c38ce24d6351f625a8873960fc070d61a446

Relation: equivalent

Source `FrontierPerimeter.lean`: 208-268: The polygonal lower bound retains positive subdivision count and the injective arc domain, with half-open subarcs avoiding overlap mass.

Findings: none.

cap_upper_external_density

Source hash: 0d4d779bde08a636e8536f594b78dfa9705a31925f9d2e9248d0d62fdf5aa11a

Relation: equivalent

Source `CMVGeometry.lean`: 1329-1340: The author retains upper orientation, base height and the open parameter interval, excluding the density-one endpoints.

Findings: none.

cap_upper_weightedArea

Source hash: 2f2165b544886739ec40b854c29666e67f06c7a423bd046c04eb224a5c6ee0ce

Relation: equivalent

Source `CMVGeometry.lean`: 2334-2359: Upper orientation and base at the upper interface are retained; only the planar-null chord has the other density value.

Findings: none.

cap_upper_weightedIntegral

Source hash: 45644ff66dcb2be9e3cb6c626527848bf909f532df9be34ffa1977e24d25b8f4

Relation: equivalent

Source `CMVGeometry.lean`: 1353-1372: The weighted arc integral equals exterior density times arc length because the endpoint exceptions are parameter-null.

Findings: none.

continuous_realizedCapParam

Source hash: 439602ca8a442d23dec8f858626a27ca71e9723ce90b04b56ee6e94cd89590f1

Relation: equivalent

Source `FrontierPerimeter.lean`: 442-450: The realized cap curve is continuous by its circular representation and the continuous isometry.

Findings: none.

continuous_realizedLeftCoreParam

Source hash: 4dc1fbf3a43e8117f791c26e28bd5a2a2466716abc2324d51a2522841ccc46bf

Relation: equivalent

Source `FrontierPerimeter.lean`: 727-735: Continuity follows from the explicit trigonometric coordinates and the Euclidean coordinate transport.

Findings: none.

continuous_realizedRightCoreParam

Source hash: db06971d2e0a6e54b3ce1b2cb1ffe296cb5ed7ac07495118f10c8f5f82bfe132

Relation: equivalent

Source `FrontierPerimeter.lean`: 736-744: The right realized curve has the same continuity property under its explicit trigonometric definition.

Findings: none.

continuous_unitCircleArc

Source hash: 343bc4472a64687ca829ebbf897c060cb6c7c3cad3716259e96fb84cf091470c

Relation: equivalent

Source `FrontierPerimeter.lean`: 14-17: Continuity is unconditional in radius under the total division convention; no positive-radius hypothesis is inserted.

Findings: none.

coreArcEnd

Source hash: 7ee06e3535f2fd49529132649f006dc8125adff906e2bb1c8e325a0ca4b835a4

Relation: equivalent

Source `CMVGeometry.lean`: 1474-1476: The core arc ends at the positive radius-side-angle product, giving the stated span.

Findings: none.

coreArcStart

Source hash: 4c395d23f09d7608737696a9d4043f61933ddaffbb62953e4470c1085d950b4e

Relation: equivalent

Source `CMVGeometry.lean`: 1472-1473: The core arc starts at the negative radius-side-angle product for both side parametrizations.

Findings: none.

core_integrableOn

Source hash: ebecbc727d05931a9afed5e77aa787703ada65169aa70e3f372bec6e04bddcac

Relation: equivalent

Source `CMVGeometry.lean`: 2309-2314: The finite core volume supplies integrability over the actual core carrier for every real density.

Findings: none.

ell

Source hash: 9adb32c6bf9324acec1e791c0ec46f0e17df835ffee7b3549cab07bdc3a760c9

Relation: equivalent

Source `MorgansArcFunction.lean`: 89-91: The scalar function is defined for every real angle under total division; its geometric interpretation is correctly restricted to the admissible cap interval.

Findings: none.

euclideanJoinSet

Source hash: 96d51d86f6c32a7f1c31ecb0290be17bec48e5d48a1aa38fdfbae4457025d389

Relation: equivalent

Source `FrontierPerimeter.lean`: 1257-1259: The Euclidean junction set is exactly the homeomorphic image of the coordinate junction set.

Findings: none.

euclideanJoinSet_finite

Source hash: 9539ffd4336468eab518f7e951c9dc4db7ee2d6bd08f0fa98bbda9dcf0817f38

Relation: equivalent

Source `FrontierPerimeter.lean`: 1260-1262: The image of the finite junction set remains finite, with unrestricted real chord parameter.

Findings: none.

euclideanPairwiseOverlap

Source hash: c1559aa1737fb404d99a3376b70fbd4f0ff514e71a947a7d30b01a1ce4c1177e

Relation: equivalent

Source `FrontierPerimeter.lean`: 1591-1597: Despite the source name, the definition is generic in the underlying type; the author correctly preserves this generality.

Findings: none.

euclideanStripDensity

Source hash: 3d074997fd7c2f4c47fc25c7e8a9a39259d6fd4628b103b9b3e4da77e1679848

Relation: equivalent

Source `FrontierPerimeter.lean`: 1389-1391: The density on the Euclidean realization is the coordinate density composed with the inverse homeomorphism, preserving interface values.

Findings: none.

euclideanStripDensity_integrableOn_of_measure_ne_top

Source hash: 509b7e5c9b45ab61b5e958fce825f92bda813640616ddf5c7591ac420818c5d2

Relation: equivalent

Source `FrontierPerimeter.lean`: 1404-1411: Finite Hausdorff measure and globally bounded measurable density give integrability for every real density parameter; no set-measurability premise is added.

Findings: none.

exact_euclidean_capParam_hausdorffMeasure

Source hash: ce3a2c8ff8b380f7da5859563d117de2848052560e457ef5f88e25aa88808f43

Relation: equivalent

Source `FrontierPerimeter.lean`: 457-483: The measure concerns the actual realized parameter image, and the isometry transports the previously established circular Hausdorff length.

Findings: none.

exact_euclidean_capTrace_hausdorffMeasure

Source hash: 8a0602f46d010a0d38a019d5e1052cfe4c5d69fc3db5988eaac8c5dbcc5c5e1c

Relation: equivalent

Source `FrontierPerimeter.lean`: 1081-1088: The actual cap trace has the same Hausdorff measure as its exactly identified realized parameter image.

Findings: none.

exact_euclidean_leftArcTrace_hausdorffMeasure

Source hash: f6b756b038d2dd53ff9e214034534494d6216a6823376bb73c389a754a1ad0d9

Relation: equivalent

Source `FrontierPerimeter.lean`: 821-827: Substituting the exact left trace-image equality gives the same Hausdorff measure as the parameter image.

Findings: none.

exact_euclidean_leftCoreParam_hausdorffMeasure

Source hash: e0211c9e8b0941bc0bbeb3bd977afe28d78688025726aac98a4d738bf7f180e9

Relation: equivalent

Source `FrontierPerimeter.lean`: 745-772: The formal difference of final and initial parameters is the stated twice-radius-side-angle length in the extended nonnegative reals.

Findings: none.

exact_euclidean_rightArcTrace_hausdorffMeasure

Source hash: 47e43c0cab0cb015904a2fe672eeec0720a25537d058b25dd2b767b05bcf1d39

Relation: equivalent

Source `FrontierPerimeter.lean`: 828-834: Substituting the right trace-image equality gives its exact Euclidean length, not a separately assigned value.

Findings: none.

exact_euclidean_rightCoreParam_hausdorffMeasure

Source hash: 5b9d17b9e5921b7fba027be4ebfa3845e68c3c43a92a67ce9d4854bf1b244db0

Relation: equivalent

Source `FrontierPerimeter.lean`: 773-800: The right parameter image has the same exact span measure, with the admissible closed side-angle endpoint included.

Findings: none.

exact_euclidean_segmentCarrier_hausdorffMeasure

Source hash: f2e6ec79942c7df8c258fab2922138cb782506dbaa0184c81a94b12e6e9b9896

Relation: equivalent

Source `FrontierPerimeter.lean`: 937-946: The affine-segment Hausdorff measure is its endpoint distance, equal to the specified positive scalar length.

Findings: none.

fourArc_euclidean_frontier

Source hash: 748ff0ec3e12377aa952eabaeafb59718df5bc069c8651bdcd45a9c687a1d41d

Relation: equivalent

Source `FrontierPerimeter.lean`: 1338-1348: The homeomorphism transports the complete coordinate frontier into exactly four Euclidean trace images, without adding chords.

Findings: none.

fourArc_frontier_weightedPerimeter_eq

Source hash: e95ee1864c7bc7d9bc3a3ec2659625a510c6cefccdafe0cd2efab0e2d93e1067

Relation: equivalent

Source `FrontierPerimeter.lean`: 1794-1844: The proof unfolds the canonical frontier measure, transports the integral, decomposes the exact frontier and evaluates its arc integrals; it does not infer geometry from a scalar sum.

Findings: none.

fourArc_pairwiseOverlap_h1_null

Source hash: 01eb58d468da4075ed5a4b3c4808bf8deddca98be0f05ef720c74c5a64fd9f4f

Relation: equivalent

Source `FrontierPerimeter.lean`: 1269-1279: The overlap set is contained in the null junction set, giving the exact measure-zero assertion needed for additivity.

Findings: none.

fourArc_pairwiseOverlap_subset_joinSet

Source hash: eea91f9cc72f6aabac972107df3b4a4acd7547fe5fa392689f59302f202356a4

Relation: equivalent

Source `FrontierPerimeter.lean`: 1142-1195: The pairwise overlap inclusion confines every shared arc point to one of the four interface junctions.

Findings: none.

hausdorffMeasure_image_Ico_eq_Icc

Source hash: 1c850c9b7ca6e169808ebfb87cac4902c44102258f4c3ba0c9b6433987f2a5de

Relation: equivalent

Source `FrontierPerimeter.lean`: 142-153: The extended metric setting and Borel structure are retained; deleting the endpoint changes Hausdorff measure by zero even for a degenerate interval.

Findings: none.

hausdorffMeasure_one_euclideanJoinSet

Source hash: 7ebe8fae03ad531268ff769afa248cd1067ee674eba1a7f8bc77f36d3ab49ebc

Relation: equivalent

Source `FrontierPerimeter.lean`: 1263-1268: Finite junction sets have zero positive-dimensional Hausdorff measure in the Euclidean realization.

Findings: none.

hausdorffMeasure_unitCircleArc_image_eq

Source hash: f4746d8c9b2c6ef1416bccca1f814590cdef08c393e10b7189ff167eb0402ab3

Relation: equivalent

Source `FrontierPerimeter.lean`: 294-311: The Lipschitz upper bound and limiting polygonal lower bound identify actual Hausdorff length on the strict sub-full-circle domain.

Findings: none.

hausdorffMeasure_unitCircleArc_image_eq_of_span_lt_two_pi

Source hash: a5a68294ffad2a36064599c08079161b28ea03171c66437e00e8f24eba632c26

Relation: equivalent

Source `FrontierPerimeter.lean`: 350-389: The strict positive span and strict sub-circumference bound are retained, so the measured image has length equal to the span without multiple traversal.

Findings: none.

hausdorffMeasure_unitCircleArc_image_le

Source hash: 07ee598762aa4a7964c4db549c22833ebf32ee2d325dcc190dd9a72212015319

Relation: equivalent

Source `FrontierPerimeter.lean`: 93-104: The upper bound permits reversed interval endpoints; the author's nonnegative-part convention matches the extended-real embedding.

Findings: none.

image_Ico_eq_image_Icc_diff_endpoint

Source hash: c4fb0e64c0405d8f9fa3cb2429d9a9bc01ac95be8aba1e08560d1aeae3f939e9

Relation: equivalent

Source `FrontierPerimeter.lean`: 124-141: Injectivity is assumed only on a set containing the closed interval, and the half-open image removes exactly the final image point.

Findings: none.

integral_four_union

Source hash: 8b3fbc02eae74d84636263a1b0bedcfff91c58bcc3410ac92807b51d54cc9ea4

Relation: equivalent

Source `FrontierPerimeter.lean`: 1562-1590: All six almost-disjointness premises, measurability of only the last three sets and integrability on all four are faithfully retained.

Findings: none.

integral_semicircle_scaled

Source hash: fe66bbeed922271b7cf0e4043f58e6b4f6651310c0d02aa776383ac42b18240d

Relation: equivalent

Source `CMVGeometry.lean`: 1759-1791: Positive radius scaling and the factor two for full horizontal section width give the stated segment area.

Findings: none.

integral_semicircle_translated

Source hash: 06721e5dc911d180205234ff69b820e369a8d82b66f142e17c0a1377b9b27f8d

Relation: equivalent

Source `CMVGeometry.lean`: 1792-1800: The vertical translation shifts both endpoints and the radicand by the same real center coordinate.

Findings: none.

integral_sqrt_one_sub_sq_from_cos

Source hash: e4a969c6c1b67c6f267db8fb9582c7cd96174fa24489265c840c646c3085a52a

Relation: equivalent

Source `CMVGeometry.lean`: 1729-1758: The integral formula includes both limiting cap angles and the major-segment range; the integration orientation and factor one half agree.

Findings: none.

intervalPushforward

Source hash: 366d70f2307ddf912f8fe8519c486284fdacdab2c9bb76f371f9a65dd8ee807d

Relation: equivalent

Source `CMVGeometry.lean`: 1200-1202: The parameter measure is Lebesgue measure restricted to the closed interval and then pushed forward, exactly as described.

Findings: none.

isometry_capComplexIsometryMap

Source hash: 5d4136606591a26f6abf24f8ba286ddba7154e77e9f96c81673196c1854009b0

Relation: equivalent

Source `FrontierPerimeter.lean`: 398-408: The map preserves Euclidean squared distances in both orientation branches, exactly the asserted isometry.

Findings: none.

isometry_leftCoreComplexIsometryMap

Source hash: a61c4ab4c222e88c43376bf46386ccf7697c506b7a23f81458ff102288a9e14a

Relation: equivalent

Source `FrontierPerimeter.lean`: 685-695: The left-center translation and horizontal reflection preserve the Euclidean metric.

Findings: none.

isometry_rightCoreComplexIsometryMap

Source hash: 2857926ac4225e60a51ede217224fdFDA63879d202e0be14411ea5c13b2fc6bf

Relation: equivalent

Source `FrontierPerimeter.lean`: 696-704: The right-center translation preserves Euclidean distances, giving the stated isometry.

Findings: none.

isometry_unitCircleArcRotation

Source hash: 0390b164e94132c53e908a7960b31f68dc786e7272a9a9f6482fa79cade88362

Relation: equivalent

Source `FrontierPerimeter.lean`: 315-322: Unit complex modulus makes the map an isometry for all real parameter values, including the total zero-radius convention.

Findings: none.

joinSet

Source hash: 613dcfda543c5255d80175237d7db38b3386ce55bc3215682252569ce26c42d5

Relation: equivalent

Source `FrontierPerimeter.lean`: 1115-1117: All four signed half-chord junction coordinates are retained, with no positivity restriction on the defining real parameter.

Findings: none.

joinSet_finite

Source hash: c95df1795c640672a88890bc658373f829154b7fa3a2361c967c14e17f22b551

Relation: equivalent

Source `FrontierPerimeter.lean`: 1253-1256: The explicit finite set has at most four points, including possible repetitions for arbitrary real chord parameters.

Findings: none.

leftCoreBoundaryMeasure

Source hash: e243bfd8c440c79be3e701f3e1ba7f0b4af716d3ad6d1c3de2eb88023b1a30c8

Relation: equivalent

Source `CMVGeometry.lean`: 1487-1490: The left auxiliary boundary measure is exactly the pushforward of the specified closed arclength interval.

Findings: none.

leftCoreBoundaryMeasure_integrable

Source hash: efdd79d96b8f43a7c478608fb7242070e102fd586af27d2826a566f07542e0a9

Relation: equivalent

Source `CMVGeometry.lean`: 2275-2281: The left arc's finite parameter interval and measurable parametrization yield integrability for every real density.

Findings: none.

leftCoreComplexIsometryMap

Source hash: c0385e604fc62ea1f5b752f17a4538ba2edeedf20e52cbfb6848b3b966b2ab57

Relation: equivalent

Source `FrontierPerimeter.lean`: 677-680: The map uses horizontal reflection and left-center translation while leaving the imaginary coordinate as height.

Findings: none.

leftCoreParam

Source hash: 80f940fe52c510e69a283605275f10b49a7ca369836a80a40ece896dbbf606db

Relation: equivalent

Source `CMVGeometry.lean`: 1477-1481: The minus cosine displacement selects the left exposed arc; the sine height and radius scaling agree.

Findings: none.

leftCoreParam_image_Icc

Source hash: 01c2af95d70e04d0bac530bc681fda34196e33e841f34e26466b4a3a9c6ecd18

Relation: equivalent

Source `FrontierPerimeter.lean`: 517-596: The image theorem is exact on the closed left arc interval; reconstruction uses the side condition to choose the correct cosine branch.

Findings: none.

leftCore_strip_density

Source hash: 3222d64a655ce3af15a638606821b923e89c31ae75558031c08763aeb76b72b6

Relation: equivalent

Source `CMVGeometry.lean`: 1662-1667: The density is one throughout the closed left arc interval, including both interface endpoints.

Findings: none.

leftCore_weightedIntegral

Source hash: e333f475432d30c40456ab56a9f5187cebdb61288c7e07d5eec293804ac77f49

Relation: equivalent

Source `CMVGeometry.lean`: 1675-1694: The density-one integral equals the parameter span, which the curvature-angle identity converts to the stated normalized length.

Findings: none.

lintegral_semicircle_translated

Source hash: 3c8e90cd635d133e832b64fe2731d46db4f68d0a1a341d292df3904cf66e67e8

Relation: equivalent

Source `CMVGeometry.lean`: 1801-1819: The nonnegative integral and extended-real embedding are distinguished from the preceding ordinary interval integral, with the same angle domain.

Findings: none.

lipschitzWith_unitCircleArc

Source hash: 6ddc026dd017d7b20d15e385a48bc34f5bcbfcad3796885aedaeb1fd06730313

Relation: equivalent

Source `FrontierPerimeter.lean`: 33-54: For positive radius the parametrization is one-Lipschitz in its arclength parameter, exactly the claimed metric bound.

Findings: none.

measurableSet_euclidean_capTrace

Source hash: 345ff49d99187e89c9b6a7558ffa41e164108dfd32b3dc525b699de014a17b5a

Relation: equivalent

Source `FrontierPerimeter.lean`: 1104-1109: The full cap trace is a compact realized parameter image for every admissible half-angle, including major caps.

Findings: none.

measurableSet_euclidean_leftArcTrace

Source hash: a31ed55013e95ae690ffcc5bab6587c2933302f4970b8b673a167e6c53e1b7f8

Relation: equivalent

Source `FrontierPerimeter.lean`: 1089-1093: The left Euclidean trace is the continuous image of a compact interval and hence measurable.

Findings: none.

measurableSet_euclidean_rightArcTrace

Source hash: 771198ca90f5054f065ccd16a46e41b5fce0886173f7b3c654e6b61824be845e

Relation: equivalent

Source `FrontierPerimeter.lean`: 1094-1098: The right Euclidean trace is likewise a compact parameter image and therefore measurable.

Findings: none.

measurable_stripDensity

Source hash: b6fc908d8bba478440a5c35f7c75a94366dc691bdf7d2125848e2a8f512284af

Relation: equivalent

Source `CMVGeometry.lean`: 1234-1240: The closed-strip branch set is measurable and both density branches are constants for every real parameter.

Findings: none.

monotone_capPartitionPoint

Source hash: 62e77199a2eb7f24e5f93307536675c49f368eab2d84be96574f544f2152a857

Relation: equivalent

Source `FrontierPerimeter.lean`: 167-174: Positive radius, half-angle and subdivision count give a nondecreasing sequence on all natural indices, as stated.

Findings: none.

ofReal_dist_le_hausdorffMeasure_image_Icc

Source hash: 272ec4c298bc0f3582f8dd04d91ee8113b10ec4ec1c37538e792ec5391cc0936

Relation: equivalent

Source `FrontierPerimeter.lean`: 105-123: The metric and Borel-space assumptions, continuity and non-strict endpoint order are all preserved in the endpoint-distance lower bound.

Findings: none.

pairwiseOverlap

Source hash: f2051dcd764bb55c1f4c980a58c56ba3f51d928d6141df3262dbdd919f1b5fbd

Relation: equivalent

Source `FrontierPerimeter.lean`: 1110-1114: The formal set is exactly the union of the six distinct pairwise intersections of four planar sets.

Findings: none.

pairwise_aedisjoint_of_overlap_null

Source hash: 4b46ea8bfc070cb38c17e38f03c2486a1eb119281a4706f7a55c5da843e21d49

Relation: equivalent

Source `FrontierPerimeter.lean`: 1598-1616: Nullity of the union implies nullity of each of its six intersections and hence the six almost-disjointness conclusions.

Findings: none.

planeEuclideanHomeomorph

Source hash: 0cae97f48ab4c061d7f08a907381bf52e9560477df33ffb993c23c2cbbf11049

Relation: equivalent

Source `CMVGeometry.lean`: 38-42: The defined homeomorphism preserves coordinates and topology; it is not incorrectly claimed to preserve the default product metric.

Findings: none.

realizedCapParam

Source hash: 95b1661a30d7b5d3f17854c6eabbe1a4da9d944d9ad542e596607c6cb4fd12b0

Relation: equivalent

Source `FrontierPerimeter.lean`: 430-433: The definition composes the coordinate parametrization with the Euclidean homeomorphism without changing point coordinates.

Findings: none.

realizedCapParam_eq

Source hash: 9c421cce21d93c49e9d419ecc30c8d31de995d391da25f034d29cf3f0bd85599

Relation: equivalent

Source `FrontierPerimeter.lean`: 434-441: Expansion of real and imaginary coordinates gives exact equality of the realized parametrization and the isometric image of the standard circle.

Findings: none.

realizedCapParam_image_Icc

Source hash: a37b40496e1e5c2ae0b778fe01e7fee7a91557ce9473bf6e0b7755afebbd941b

Relation: equivalent

Source `FrontierPerimeter.lean`: 1071-1080: The homeomorphism transports the exact cap image equality to the Euclidean trace used for length.

Findings: none.

realizedLeftCoreParam

Source hash: 854e6d1ea17e2b2d6576f273610fd43051ac61c0bb7e75c185b3d3aa821f7da4

Relation: equivalent

Source `FrontierPerimeter.lean`: 705-708: The left realized curve is the coordinate parametrization followed by the Euclidean homeomorphism.

Findings: none.

realizedLeftCoreParam_eq

Source hash: e2c0026193d6fa0dd58c252b75c6df1a108616049c60130059ddb93311c0b2a0

Relation: equivalent

Source `FrontierPerimeter.lean`: 713-719: The left realized curve is exactly the left-center isometric image of the standard circle at the same arclength parameter.

Findings: none.

realizedLeftCoreParam_image_Icc

Source hash: 43b7188db3ec73adee2747ff0dc65e5368084e5e17965fa7fdaa8f3c9301ccd7

Relation: equivalent

Source `FrontierPerimeter.lean`: 801-810: The coordinate image theorem is transported through the homeomorphism, giving equality with the Euclidean left trace.

Findings: none.

realizedRightCoreParam

Source hash: 4f8de565538cdbe799e2faf0a2d6452b4030fcfacd5b43a7347015b8e46924a9

Relation: equivalent

Source `FrontierPerimeter.lean`: 709-712: The right realized curve uses the same coordinate transport and retains the right parametrization.

Findings: none.

realizedRightCoreParam_eq

Source hash: 4aab5f14bc5465771d4021be2ff0ba59e74b69eeaba0b9a9c834bd0bf72ed625

Relation: equivalent

Source `FrontierPerimeter.lean`: 720-726: The right realized curve is exactly the right-center isometric image of the standard circle at the same parameter.

Findings: none.

realizedRightCoreParam_image_Icc

Source hash: e28432904f0d4f1c849ed6ed56e30bdae00185a8e05f01ccceab8ef89d16f29b

Relation: equivalent

Source `FrontierPerimeter.lean`: 811-820: The corresponding coordinate transport identifies the realized right image with the Euclidean right trace.

Findings: none.

realizedSegmentParam

Source hash: 4c1728c18c2df9a1da871e357fce4f6439c6307fe654c71c273fe06b8caaa0fd

Relation: equivalent

Source `FrontierPerimeter.lean`: 864-867: The realized segment parametrization changes only the metric representation, preserving its affine coordinates.

Findings: none.

realizedSegmentParam_image_affineSegment

Source hash: c3be9a5bc2d447d5c5665690a46411881b4f0933eb5c3cf50deaa41a88946c72

Relation: equivalent

Source `FrontierPerimeter.lean`: 873-912: The closed parameter image is exactly the affine segment between the realized endpoints, with no missing endpoint or open-segment convention.

Findings: none.

realizedSegmentParam_image_carrier

Source hash: fc21a65690cb2d00a27cb6aebde20046f1dffbc15b7f30d3caf58ac1aefd1725

Relation: equivalent

Source `FrontierPerimeter.lean`: 913-922: The coordinate carrier image and the realized parameter image coincide by composition with the homeomorphism.

Findings: none.

rightCoreBoundaryMeasure

Source hash: 33cbc13fa6fcd71207ac7e5dfb0579966744c8887bb1c9af1b95e6c688f55ea8

Relation: equivalent

Source `CMVGeometry.lean`: 1491-1493: The right auxiliary boundary measure uses the corresponding right parametrization and identical interval, without chord terms.

Findings: none.

rightCoreBoundaryMeasure_integrable

Source hash: 63cd5ea28e72c7cf39aef8bfd2e698b16940d7ae74acc20f074d60c55994913c

Relation: equivalent

Source `CMVGeometry.lean`: 2282-2288: The right arc has the same finite-measure integrability property with no positivity restriction on density.

Findings: none.

rightCoreComplexIsometryMap

Source hash: 24206b125ec8e6de86a054126e34b32568c5cb9c365728de42d394ff9cc0ede6

Relation: equivalent

Source `FrontierPerimeter.lean`: 681-684: The map uses right-center translation with the positive real coordinate, exactly as defined.

Findings: none.

rightCoreParam

Source hash: 38ea0d845e704b5d320adc3936a1b6a5fba52e17ce5fb320da7c20200afa9e4d

Relation: equivalent

Source `CMVGeometry.lean`: 1482-1486: The plus cosine displacement selects the right exposed arc; the same sine height interval is used.

Findings: none.

rightCoreParam_image_Icc

Source hash: 78bb53ade5c270538e82ae914ea314e16857498de0ee3dfc73dde7468e65c773

Relation: equivalent

Source `FrontierPerimeter.lean`: 597-676: The right arc has the corresponding exact closed-interval image, with the opposite horizontal side selection.

Findings: none.

rightCore_strip_density

Source hash: d879e18591ac7ff00f0a228fad33d85d1e290fb3c77820aaf80c672ec9411662

Relation: equivalent

Source `CMVGeometry.lean`: 1668-1674: The density-one conclusion includes the entire closed right arc interval and needs no density contrast assumption.

Findings: none.

rightCore_weightedIntegral

Source hash: 12c65212e5417b38acb8560f789f61c31ede351c3de6efe2b847438f7d6cf718

Relation: equivalent

Source `CMVGeometry.lean`: 1695-1714: The same arclength calculation yields the right arc contribution with the correct factor two.

Findings: none.

segmentEnd

Source hash: 161abc8655ce59212c511d028e178416823adf64c523c73dae6b889ae03ed1f4

Relation: equivalent

Source `CMVGeometry.lean`: 1400-1402: The final segment parameter is the positive half-length, exactly matching the right endpoint.

Findings: none.

segmentEndpoint_edist

Source hash: 667e4fe55a19b902ccd7cc29782bd67f737d862a1bddf625b46565ed0a83a1cd

Relation: equivalent

Source `FrontierPerimeter.lean`: 923-936: The equal endpoint heights leave exactly the positive horizontal separation as extended Euclidean distance.

Findings: none.

segmentParam

Source hash: ba8ed6bbc5cd19b9c405036bea7c221ae24e5c76ea0f6ef5cc2d62c327ae1dc5

Relation: equivalent

Source `CMVGeometry.lean`: 1395-1397: The affine parametrization is horizontal displacement from the midpoint at fixed base height.

Findings: none.

segmentParam_image_Icc

Source hash: 78cfdac8328c8d0308b83fec45a7d72c3a0a830edb6a3f41f9bd2da54db0f76

Relation: equivalent

Source `FrontierPerimeter.lean`: 835-863: The affine segment parametrization has exactly the closed interval's carrier as its image, with both endpoints present.

Findings: none.

segmentStart

Source hash: e90adf884523aab9d2e3af449adbc9d655d375051a612c0b4970a212a5dca61f

Relation: equivalent

Source `CMVGeometry.lean`: 1398-1399: The initial segment parameter is the negative half-length, exactly matching the left endpoint.

Findings: none.

stripDensity_integrableOn_of_volume_ne_top

Source hash: d26957a576fdcc692079b92ac5054cbbeda2a914af15d77e6d70ad1d418d20fd

Relation: equivalent

Source `CMVGeometry.lean`: 2289-2302: Finite ambient volume is the only set hypothesis; the statement does not add a measurability assumption absent from the formal theorem.

Findings: none.

stripDensity_integrable_intervalPushforward

Source hash: 6315c9471ea2031d23e0cb4552b23f2812fe90938358c371a246f04d7f2d8095

Relation: equivalent

Source `CMVGeometry.lean`: 2243-2259: The measurable-curve hypothesis and arbitrary real interval endpoints are retained; finite parameter measure suffices for bounded-density integrability.

Findings: none.

tendsto_cap_polygon_length

Source hash: 32d7803e3d290677cdac6cd7bd331241d0a1e99888d3a8665e326de2ca241a11

Relation: equivalent

Source `FrontierPerimeter.lean`: 269-293: Only the half-angle is assumed positive; the author preserves the all-real radius parameter in the small-angle limit.

Findings: none.

unitCircleArc

Source hash: 6865b0b68c07403ee39252efa35feb9b50850a4644c2bf915fb506723c7a07c7

Relation: equivalent

Source `FrontierPerimeter.lean`: 11-13: The complex exponential defines a circle of arbitrary real radius parameter, not specifically the unit circle suggested by the source name.

Findings: none.

unitCircleArcRotation

Source hash: cfdc45400ed0d08a69b42a215854d1f81999a500771f48a7f18476aafefe48ef

Relation: equivalent

Source `FrontierPerimeter.lean`: 312-314: The rotation is multiplication by the stated unit complex exponential and is total in both real parameters.

Findings: none.

unitCircleArcRotation_apply

Source hash: dcedc37c9428c0a722c5bf130023e2389a83c7faa4036432847247c3401f6b9b

Relation: equivalent

Source `FrontierPerimeter.lean`: 323-331: The exponential addition law shifts the arclength parameter by the stated amount; the formal nonzero-radius premise is retained.

Findings: none.

unitCircleArcRotation_image_Icc

Source hash: 6bbfe5bd6af2f20e278e2e30ff5adc9fe7c0761359113d169ea5baae1f20d6a2

Relation: equivalent

Source `FrontierPerimeter.lean`: 332-349: The image equality translates both interval endpoints and allows empty reversed intervals, with the same nonzero-radius premise.

Findings: none.

unitCircleArc_dist

Source hash: 1c583965a8663dcd3d0483716b4801199d867b2ff4cbe96f04d4cbeb57ff9c08

Relation: equivalent

Source `FrontierPerimeter.lean`: 55-66: The chord-distance formula includes the absolute value and half-angle factor, with a positive radius outside the absolute value.

Findings: none.

unitCircleArc_eq

Source hash: 9342d6a8ed8b7dcd1fbce36251ddf6b271aeea6ec0298f5d467e178c8fbf9cec

Relation: equivalent

Source `FrontierPerimeter.lean`: 409-415: Euler's identity yields the stated sine and cosine coordinates for every real radius and parameter.

Findings: none.

unitCircleArc_im

Source hash: 0666e8d860656cd1bc24ade42af99acdafd9034497a0ba0f27acf44cddcd77e5

Relation: equivalent

Source `FrontierPerimeter.lean`: 423-429: The imaginary coordinate is the radius-scaled sine of the normalized parameter, including total values.

Findings: none.

unitCircleArc_injOn

Source hash: ceabbbb1bd00ed7b3d5a248b0d9d8b4d41f8f1fc0af8194bd56a8092deed637e9

Relation: equivalent

Source `FrontierPerimeter.lean`: 67-92: The strict half-angle upper bound prevents a complete repeated traversal, giving injectivity on the closed symmetric parameter interval.

Findings: none.

unitCircleArc_re

Source hash: 238aa74976048276575d89eb69760ceb968a224b5358a1775e61818ec60df916

Relation: equivalent

Source `FrontierPerimeter.lean`: 416-422: The real coordinate is the radius-scaled cosine of the normalized parameter, including total values.

Findings: none.

unitCircleArc_sub_factor

Source hash: d6c59e7f498e1352e21969a1c8084fa6600038a792db1172f4e5bd3b2836eece

Relation: equivalent

Source `FrontierPerimeter.lean`: 18-32: The exponential factorization and nonzero-radius premise are both retained, with the same difference orientation.

Findings: none.

upper_cap_integral

Source hash: 447beffafa9faf9c0047e982aa709b3b243852770c1fde70cc3c6676215458bf

Relation: equivalent

Source `FrontierPerimeter.lean`: 1545-1552: Upper orientation and the upper interface are retained; actual trace integration uses exterior density except at the two length-null endpoints.

Findings: none.

volume_horizontalLine

Source hash: ea745db5e4b12e1d05fe0934ca3c81eb24b84ea87dd522e3aa6824c86d1a2dce

Relation: equivalent

Source `CMVGeometry.lean`: 2029-2036: The zero measure is planar Lebesgue measure; the explanation correctly avoids treating the line as length-null.

Findings: none.

volume_lower_circular_segment

Source hash: 2301e25d7ced8d1d9e2c23a9b63e02ba7c6e8f9ea615112d4c41d15aeb777045

Relation: equivalent

Source `CMVGeometry.lean`: 1945-1995: The lower half-plane cutoff has the reflected sign, and the formula preserves volume under vertical reflection.

Findings: none.

volume_upper_circular_segment

Source hash: d23f6efcc432aaa58959c64bfa3f6f589362c33546093657b1717ec5dbfb1b34

Relation: equivalent

Source `CMVGeometry.lean`: 1820-1944: The disk/half-plane volume formula is valid on the entire closed cap-angle interval; it is not limited to minor segments.

Findings: none.

volume_verticalLine

Source hash: b32e9d96eca7ded1e8b5354b6d117e61bcda7b34c1fe056587cd14e1871da61b

Relation: equivalent

Source `CMVGeometry.lean`: 2037-2046: The vertical line is planar-null for every real horizontal coordinate, supporting area decomposition only.

Findings: none.

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